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A dynamic bactofilin cytoskeleton recruits an M23 endopeptidase to control bacterial morphogenesis

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Manuel Osorio-Valeriano, Emöke Cserti, Sebastian Pöhl, Rogelio Hernández-Tamayo, Jacob Biboy, Patrick Sobetzko, Waldemar Vollmer, Peter L. Graumann, Martin Thanbichler

Department of Biology, University of Marburg, Marburg, Germany • Max Planck Institute for Terrestrial Microbiology, Marburg, Germany • Department of Chemistry, University of Marburg, Marburg, Germany • Center for Synthetic Microbiology (SYNMIKRO), Marburg, Germany • Centre for Bacterial Cell Biology, Institute for Cell and Molecular Biosciences, Newcastle University, Newcastle upon Tyne, United Kingdom

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Abstract

Bactofilins have emerged as a widespread family of cytoskeletal proteins with important roles in bacterial morphogenesis, but their precise mode of action is still incompletely understood. Here, we identify the bactofilin cytoskeleton as a key regulator of cell growth in the stalked budding alphaproteobacterium *Hyphomonas neptunium*. We show that in this species, the lack of bactofilins causes severe morphological defects, resulting from unconstrained growth of the stalk and bud compartments. In line with this finding, bactofilin polymers localize dynamically to the stalk base and then to the incipient bud neck prior to the onset of bud formation, suggesting that they act as a barrier that retains the cell wall biosynthetic machinery in the respective growth zones. Notably, in a broad range of species, bactofilin genes lie adjacent to genes encoding cell wall hydrolases of the M23 peptidase family. We show that the corresponding *H. neptunium* endopeptidase, LmdC, is a bitopic membrane protein with peptidoglycan hydrolase activity that colocalizes with the bactofilin cytoskeleton, dependent on a direct interaction of its cytoplasmic tail with the bactofilin cytoskeleton. A functional association of bactofilins with M23 peptidases is further verified by studies of the spiral-shaped alphaproteobacterium *Rhodospirillum rubrum*, whose bactofilin and LmdC homologs colocalize at the inner cell curvature, forming a complex that modulates the degree of cell helicity. These findings indicate that bactofilins and M23 peptidases form a conserved functional module that is critical for cell shape determination in morphologically complex bacteria.

eLife assessment

The findings in this manuscript have **important** theoretical and practical implications within the fields of bacterial morphogenesis, peptidoglycan, and cytoskeletal function. The evidence presented is **convincing** and uses validated methodology in line with current state-of-the-art. This manuscript will be of interest to cell biologists studying the cytoskeleton and bacterial morphogenesis.

Introduction

Bacteria come in a variety of different cell shapes, which can be further modified by the formation of cellular extensions such as branches or stalks (Kysela et al., 2016; Yang et al., 2016). Their morphology can change as a function of the cell cycle or in response to environmental cues to ensure optimal fitness in the given ecological niche or growth conditions (van Teeseling et al., 2017; Yang et al., 2016). In the vast majority of species, cell shape is determined by the peptidoglycan cell wall, a complex macromolecule composed of glycan chains that are crosslinked by short peptides (Typas et al., 2012; Vollmer et al., 2008). The synthesis of this mesh-like structure is achieved by an array of synthetic and lytic enzymes that are typically combined into multi-protein complexes and associated with regulatory factors and cytoskeletal elements to facilitate the coordination and spatiotemporal regulation of their activities (Egan et al., 2020; Rohs and Bernhardt, 2021).

Basic spherical, rod-like and hyphal shapes are generated by the combined action of the cell division and cell elongation machinery (Margolin, 2009; Rohs and Bernhardt, 2021). In most bacteria, cell division is executed by the divisome, which is organized by the tubulin homolog FtsZ and mediates cell constriction and the synthesis of the new cell poles prior to cytokinesis (McQuillen and Xiao, 2020). Cell elongation, by contrast, can be achieved by various types of cell wall-biosynthetic complexes, including the so-called elongasome, which is organized by the actin homolog MreB and mediates the dispersed incorporation of new cell wall material along the lateral cell walls (Shi et al., 2018), or different cell pole-associated complexes that promote polar growth of the cell body (Brown et al., 2011). More complex cell shapes are generated with the help of accessory systems that either modulate the activity of the generic cell elongation machinery or have peptidoglycan biosynthetic activity on their own, thereby locally modifying the structure of the peptidoglycan layer (Taylor et al., 2019).

A particularly widespread family of cytoskeletal proteins implicated in cell shape modification are the bactofilins. They are characterized by a conserved central Bactofilin A/B domain (InterPro ID: IPR007607; (Paysan-Lafosse et al., 2022)) with a barrel-like β -helical fold that is typically flanked by short disordered terminal regions (Kühn et al., 2010; Shi et al., 2015; Vasa et al., 2015). Bactofilins polymerize spontaneously without the need for nucleotide cofactors (Koch et al., 2011; Kühn et al., 2010), driven by head-to-head and tail-to-tail interactions between the core domains of neighboring molecules (Deng et al., 2019). Lateral interactions between individual protofilaments can then give rise to higher-order assemblies, such as bundles or two-dimensional sheets (Kühn et al., 2010; Vasa et al., 2015; Zuckerman et al., 2015). Previous work has suggested that bactofilin polymers typically associate with the inner face of the cyto-plasmic membrane and localize to regions of high membrane curvature (Caccamo et al., 2020; Hay et al., 1999; Kühn et al., 2010; Lin et al., 2017; Taylor et al., 2020). These structures have been co-opted as localization determinants and assembly platforms by several different morphogenetic systems.

For instance, bactofilin homologs were reported to contribute to rod-shape maintenance in *Myxococcus xanthus* (Koch et al., 2011), the establishment of helical cell shape in the human pathogen *Helicobacter pylori* (Sycuro et al., 2010; Taylor et al., 2020) as well as the modulation of cell helicity in the spiral-shaped bacterium *Leptospira biflexa* (Jackson et al., 2018). Apart from modulating general cell shape, they were found to have an important role in the formation of cellular extensions known as stalks, which are wide-spread among alphaproteobacterial species (Wagner and Brun, 2007). Stalks are elongated protrusions of the cell envelope that are filled with a thin thread of cytoplasm and grow through zonal incorporation of cell wall material at their base (Aaron et al., 2007; Randich and Brun, 2015). In *Caulobacter crescentus* and *Asticcacaulis biprosthecum*, bactofilin polymers were shown to localize to the stalk base to direct proper stalk formation. In *C. crescentus*, they recruit a cell wall synthase that contributes to stalk elongation, with their absence leading to a

reduction in stalk length (Kühn et al., 2010). In *A. biprosthicum*, by contrast, bactofilin acts as a central topological regulator that is required to efficiently initiate stalk formation and limit peptidoglycan biosynthesis to the stalk base. Its absence leads to the development of pseudostalks, which are much shorter and wider than normal stalks and irregularly shaped, likely due to unrestrained peptidoglycan biosynthesis through the entire stalk envelope (Caccamo et al., 2020).

While stalks are often accessory structures with highly specialized functions (Klein et al., 2013; Persat et al., 2014; Wagner and Brun, 2007), stalked budding bacteria such as *Hyphomonas neptunium* and other members of the *Hyphomonadaceae* and *Hyphomicrobiaceae* use them as integral parts of the cell with key roles in cell growth and division (Moore, 1981). *H. neptunium* has a biphasic life cycle (Wali et al., 1980), in which a non-replicative, motile swimmer cell sheds its single polar flagellum and differentiates into a replicative, sessile stalked cell (Figure 1—figure supplement 1). Unlike most widely studied model species, it does not divide by binary fission but instead produces new offspring through the formation of buds at the tip of the stalk. As the terminal stalk segment gradually dilates, a flagellum is formed at the pole opposite the stalk. After DNA replication and translocation of one of the sister chromosomes through the stalk into the bud compartment (Jung et al., 2019), cytokinesis occurs at the bud neck, releasing a new swimmer cell. While the newborn cell first needs to differentiate into stalked cells to start replication, the stalked mother cell restores the stalk and then immediately re-enters the next budding cycle (Jung et al., 2019; Wali et al., 1980). The developmental program of *H. neptunium* involves several switches in the pattern of peptidoglycan biosynthesis (Cserti et al., 2017). After birth, swimmer cells increase in size by dispersed incorporation of new cell wall material throughout the cell body. Stalk formation is then achieved by zonal growth at the stalk base, followed by localized dispersed growth of the stalk-terminal bud compartment and, finally, zonal peptidoglycan synthesis at the site of cell division. The pattern of new cell wall synthesis is similar to the localization pattern of elongasome components (Cserti et al., 2017), suggesting the involvement of this machinery in all growth phases. However, the underlying regulatory mechanisms are still unknown.

In this study, we identify the bactofilin cytoskeleton as a central player in the regulation of cell growth in *H. neptunium*. We show that in this organism bactofilin polymers localize dynamically to the stalk base and the bud neck, with their absence leading to unconstrained growth of the stalk and bud compartments, indicating a central role in the spatial regulation of cell wall biosynthesis. Interestingly, database searches reveal that in a range of different species bactofilin genes are clustered with genes for cell wall hydrolases of the M23 peptidase family, suggesting a functional connection between these two types of proteins. We find that the *H. neptunium* M23 peptidase homolog LmdC indeed consistently colocalizes with the bactofilin cytoskeleton *in vivo* and interacts directly with bactofilin *in vitro*. Studies in the spiral-shaped alpha-proteobacterium *Rhodospirillum rubrum* again reveal a close association of its bactofilin and LmdC homologs, which colocalize in regions of positive inner cell curvature and are both required to ensure proper cell helicity. Collectively, these demonstrate a conserved functional interaction between bactofilins and M23 peptidases that is important for the control of cell growth in morphologically complex bacteria.

Results

The bactofilin cytoskeleton is required for cell shape determination in *H. neptunium*

The *H. neptunium* chromosome contains two open reading frames that encode bactofilin homologs, HNE_0444 and HNE_2629 (Badger et al., 2006). Reciprocal BLAST analysis using the two bactofilins described in the close relative *C. crescentus* identified HNE_2629 as a potential ortholog of *C. crescentus* BacA (43% identity, 60% similarity). By contrast, HNE_0444 was only distantly related (<30% identity) to either of the two proteins. Based on these results, we propose to designate the two bactofilin homologs of *H. neptunium* BacA (HNE_2629) and BacD (HNE_0444), respectively. *bacA* forms a putative bicistronic operon with *lmdC*, an essential gene encoding an M23 peptidase homolog (Cserti et al., 2017) (Figure 1A). The two genes overlap by 17 base pairs, suggesting that their expression is closely coupled. *bacD*, by contrast, is not part of an operon. Both BacA and BacD display the typical architecture of bactofilins, with a central polymerization domain flanked by short non-structured N- and C-terminal regions (Figure 1B).

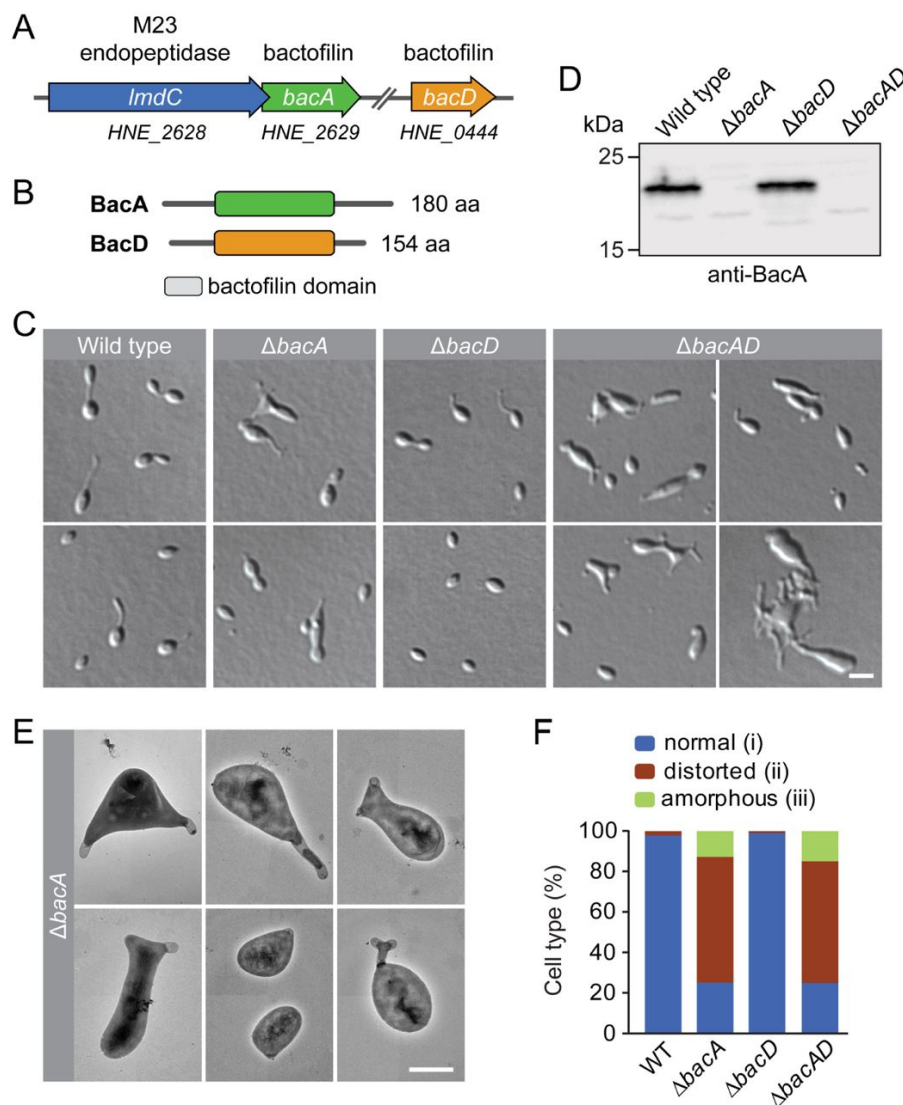


Figure 1.

Bactofilin BacA is required for proper cell morphology in *H. neptunium*.

(A) Schematic representation of the two bactofilin genes present in the *H. neptunium* genome. *bacA* lies adjacent to the M23 peptidase gene *lmdC*. Arrows indicate the direction of transcription. **(B)** Domain organization of BacA and BacD from *H. neptunium*. The bactofilin polymerization domain (colored boxes) is flanked by non-structured N- and C-terminal regions. **(C)** Morphology of *H. neptunium* bactofilin mutants. Shown are representative cells of strains LE670 (wild type), EC28 ($\Delta bacA$), EC23 ($\Delta bacD$) and EC33 ($\Delta bacAD$), imaged by differential interference contrast (DIC) microscopy. Bar: 4 μm . **(D)** Immunoblot analysis of the strains shown in panel C, performed using anti-BacA antibodies. **(E)** Transmission electron micrographs of $\Delta bacA$ cells at the early stalked-cell stage. Bar: 1 μm . **(F)** Quantification of the proportion of phenotypically abnormal stalked and budding cells in cultures of the strains analyzed in panel C (n = 100

cells per strain).

To investigate the role of bactofilins in *H. neptunium*, we generated mutant strains in which *bacA* and *bacD* were deleted either individually or in combination. Upon light and electron microscopic analysis, $\Delta bacA$ and $\Delta bacAD$ cells showed severe morphological defects, as reflected by irregularly shaped, elongated and/or oversized cells, buds directly fused with the mother cell body, branched stalks, and multiple wide protrusions that emerged from the cells in an apparently random fashion (Figures 1C–F), reminiscent of the pseudostalks reported for a bactofilin-deficient *A. biprosthecum* mutant (Caccamo et al., 2020). The wild-type phenotype could be largely restored by expressing an ectopic copy of *bacA* under the control of a copper-inducible promoter, even though BacA accumulated to lower-than-normal levels under this condition (Figure 1—figure supplement 2A–C), confirming the absence of polar effects. The deletion of *bacD*, by contrast, did not cause any obvious cell shape defects (Figures 1C,F and Figure 1—figure supplement 2D). Moreover, neither *bacD* deletion (Figure 1C,F) nor *bacD* overexpression (Figure 1—figure supplement 2C,E) had any influence on the proportion of distorted or amorphous cells in the $\Delta bacA$ background. These results demonstrate that BacA has a critical role in the regulation of cell growth in *H.*

neptunium, whereas BacD might be an auxiliary factor of so-far unknown function, similar to BacB in *C. crescentus*.

To obtain more insight into the dynamics of cell growth in the $\Delta bacA$ mutant and identify the initial phenotypic defects induced upon BacA depletion, we imaged a conditional *bacA* mutant after its transfer from permissive to restrictive conditions on an agarose pad (**Figure 2A,B**). Following cells at the swimmer-to-stalked cell transition, we observed that stalk formation initially proceeded as in the wild-type strain. However, as BacA was gradually depleted, stalk elongation ceased and the stalk structure started to widen and eventually develop multiple bulges that kept on expanding in an apparently uncontrolled manner. Thus, BacA appears to be required to maintain the polar growth zone at the stalk base, with its absence leading to unconstrained growth of the stalk cell wall. To follow the fate of bactofilin-deficient cells over a prolonged period of time, we monitored $\Delta bacAD$ cells in a flow-cell system, which ensured optimal nutrient supply throughout the course of the experiment and led to a looser packing of cells, thereby facilitating their visual analysis (**Figure 2—figure supplement 1** and **Video 1**). In this setup, cells again started growth by the formation of irregularly shaped stalks that started to branch, with branches developing either into extensive hyphal-like structures or large, amorphous compartments that may represent morphologically aberrant buds. At irregular intervals, cells divided at the junctions between hyphal and bulged cellular segments, releasing smaller amorphous fragments and, occasionally, also cells with close-to-normal morphology. Wild-type cells, by contrast, showed the usual growth pattern when cultivated under these conditions (**Video 2**). Together, these results underscore the importance of BacA in the spatiotemporal control of cell growth in *H. neptunium*.

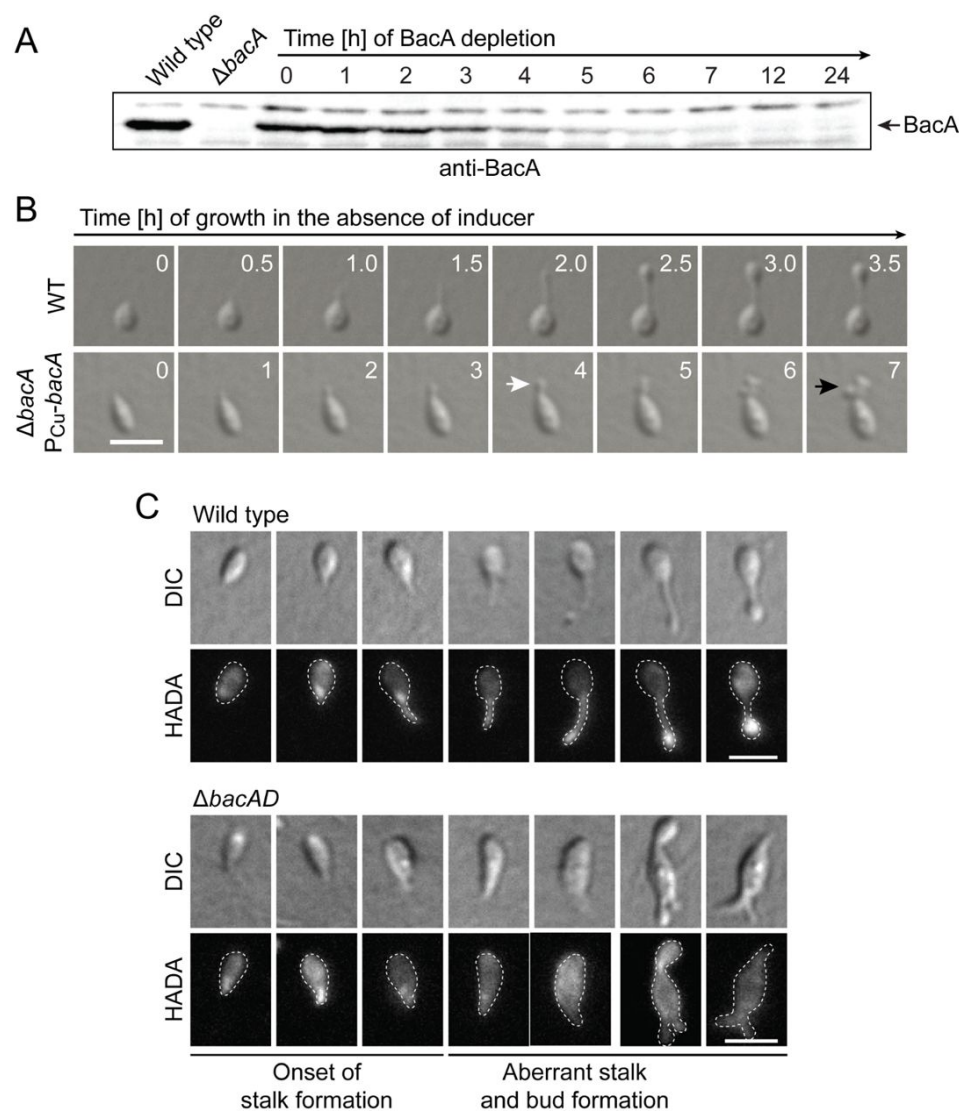


Figure 2.

Lack of BacA leads to uncontrolled growth of the stalk and bud compartments.

(A) Immunoblot showing the levels of BacA in strain EC41 ($\Delta bacA$ P_{Cu} - $bacA$) over the course of BacA depletion. Cells were grown in copper-containing medium, washed and transferred to inducer-free medium. At the indicated time points, samples were withdrawn and subjected to immunoblot analysis. Strains LE670 (wild type) and EC28 ($\Delta bacA$) were included as controls. The position of BacA is indicated. **(B)** Morphological defects induced by BacA depletion. Cells of strains LE670 (wild type) and EC41 ($\Delta bacA$ P_{Cu} - $bacA$) were grown in medium containing 0.5 mM $CuSO_4$, washed, and incubated for 6 h in inducer-free medium before they were transferred onto ASS-agarose pads lacking inducer and imaged at the indicated time points. Bar: 2 μm . **(C)** Changes in the growth pattern of *H. neptunium* in the

absence of bactofilins. Cells of strains LE670 (wild type) and EC33 ($\Delta bacAD$) were stained with the fluorescent D-amino acid HADA prior to analysis by fluorescence microscopy. Shown are representative images of cells at different developmental stages. Bars: 2 μm .

To obtain more detailed information about the dynamics of cell wall biosynthesis in the absence of bactofilins, we performed pulse-labeling studies with the fluorescent D-amino acid HADA, which is rapidly incorporated into peptidoglycan when added to the culture medium, thus marking regions of ongoing cell wall biosynthesis (Kuru et al., 2015). Wild-type cells showed the typical succession of growth modes, with dispersed growth in swimmer cells, zonal growth at the stalk base in stalked cells and localized dispersed growth in the nascent bud (Figure 2C). In the $\Delta bacAD$ background, by contrast, this switch in the growth modes was abolished. Cells with close-to-normal morphology that had just initiated stalk formation still showed a distinct fluorescent focus at the stalked pole, suggesting that the initial recruitment of the machinery responsible for stalk formation occurred in a bactofilin-independent manner. However, all other cell types, including amorphous cells with aberrant stalk- and bud-like extensions, only displayed diffuse fluorescence, which points to uncontrolled growth through dispersed incorporation of new peptidoglycan throughout the

entire cell envelope. These findings support the notion that the bactofilin cytoskeleton is required to limit cell wall biosynthesis to the different growth zones of *H. neptunium*.

Bactofilins localize dynamically to the boundaries of the *H. neptunium* growth zones

Despite their severe cell shape defects, the $\Delta bacA$ and $\Delta bacAD$ mutants showed only a moderate decrease in their apparent growth rates (**Figure 2—figure supplement 2A**), and the global composition of their peptidoglycan remained essentially unchanged (**Figure 2—figure supplement 2B** and **Dataset 1**). These findings suggested that BacA might affect cell wall composition at a local scale or act mainly by modulating the activity of the generic cell elongation machinery. To further investigate the role of bactofilins in *H. neptunium*, we first aimed to verify the ability of the proteins to assemble into polymeric scaffolds. A model of the structure of BacA generated with AlphaFold-Multimer (Evans et al., 2022) suggested that the protein adopted a β -helical fold and polymerized through head-to-head and tail-to-tail interactions among different subunits (**Figure 3A**), as revealed previously in experimental studies (Deng et al., 2019; Shi et al., 2015). Consistent with this prediction, purified BacA produced a mixture of long filaments, filament bundles and two-dimensional polymeric sheets *in vitro*, which could be readily visualized by transmission electron microscopy (**Figure 3B**). Moreover, upon heterologous co-expression in *Escherichia coli*, fluorescently tagged derivatives of BacA and BacD colocalized into extended filamentous structures that were associated with the cell envelope (**Figure 3C**). These results confirm the polymeric nature of bactofilin assemblies in *H. neptunium* and suggest that BacA and BacD interact with each other *in vivo*.

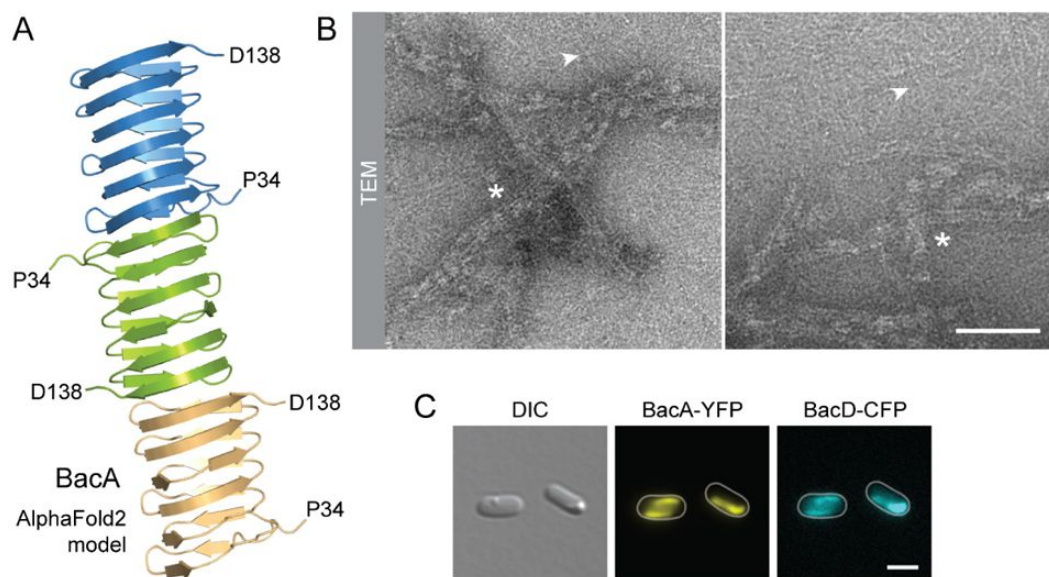


Figure 3.

BacA and BacD co-polymerize into filamentous structures.

(A) Model of a BacA trimer generated with AlphaFold-Multimer (Evans et al., 2022). Only residues P34-D138 are shown for each subunit. (B) Visualization of BacA polymers. Purified BacA-His₆ was stained with uranyl acetate and imaged by transmission electron microscopy (TEM). Arrowheads point to BacA filaments. Asterisks indicate filament bundles and sheets. Bars: 200 nm. (C) Copolymerization of BacA and BacD after heterologous coexpression in *E. coli*. Cells of *E. coli* BL21(DE3) transformed with plasmid pEC121 (P_{T7}-*bacA-eyfp* P_{T7}-*bacDecfp*) were grown in LB medium containing 5 % glucose and induced with 0.5 mM IPTG prior to imaging. Bar: 3 μm.

Next, we sought to investigate the localization dynamics of the bactofilin cytoskeleton *in vivo*. To this end, we generated strains producing fluorescently tagged BacA or BacD derivatives in place of the native proteins (Figure 4—figure supplement 1). Time-lapse microscopy analysis of cells producing a BacA-YFP fusion on agarose pads (Figure 4A) revealed that the protein was localized to the new cell pole in swimmer cells and remained associated with the stalk base during the initial phase of stalk formation. At some point, it appeared to attach to the stalk structure and then move away from the base as new cell wall material was inserted. Subsequently, stalk elongation ceased and the terminal stalk segment, delimited by the bactofilin assembly, started to swell and develop into a bud, gradually displacing BacA-YFP in the direction of the stalk base as its size increased. After cell division, both the mother cell and the newborn swimmer cells showed a fluorescent focus, indicating that the bactofilin assembly was split during cell division. To monitor the dynamics of the bactofilin cytoskeleton over multiple division cycles, we analyzed the same strain in a flow-cell system (Figure 4—figure supplement 2 and Videos 3-5). Under these conditions, the small, newborn swimmer cells were washed away immediately after cytokinesis, preventing the formation of microcolonies around the mother cell. We observed that cell division occurred at a small distance from the BacA-YFP focus, leaving a short stalk-terminal segment in between the bactofilin assembly and the stalk tip. After cytokinesis, the stalk elongated again prior to the initiation of the next budding event. Notably, BacA-YFP was only occasionally detected at the stalked pole during the stalk restoration phase, suggesting that

the bactofilin cytoskeleton no longer plays a major role in stalk growth once the stalk structure has been established. To verify the behavior observed in the time-lapse studies, we performed a population-wide analysis of the BacA-YFP localization pattern, based on snapshot images of exponentially growing cells. A demographic analysis confirmed that the protein localizes to the new cell pole in swimmer cells, remains at the stalk base in cells with short stalks and then moves to a position close to the distal end of the stalk before the onset of budding, remaining associated with the bud neck up to the point of cell division (**Figure 4B**). A very similar behavior was observed for a BacD-Venus fusion (**Figure 4C**), consistent with the notion that the two bactofilin paralogs interact. In support of this hypothesis, studies of a strain in which both proteins were fluorescently labeled showed that BacA and BacD indeed colocalized at all stages of the developmental cycle (**Figures 4D** and **Figure 4—figure supplement 3A**). In the absence of BacA, BacD-Venus still formed distinct foci, albeit at apparently random positions within the cell (**Figure 4—figure supplement 3B**). Notably, a BacA-YFP variant carrying a previously reported mutation that disrupts the polymerization interface (F130R) (Deng et al., 2019; Vasa et al., 2015; Zuckerman et al., 2015) was evenly distributed within the cell and unable to functionally replace the wild-type protein, indicating that the formation of polymeric assemblies is essential for proper BacA localization and function (**Figure 4—figure supplement 4**). Given the localization of the bactofilin assemblies to the stalk and bud boundaries and the unconstrained growth of these cellular structures in bactofilin-deficient strains, we hypothesize that the bactofilin cytoskeleton has a critical role in limiting the cell wall biosynthetic machinery to the different growth zones of *H. neptunium*.

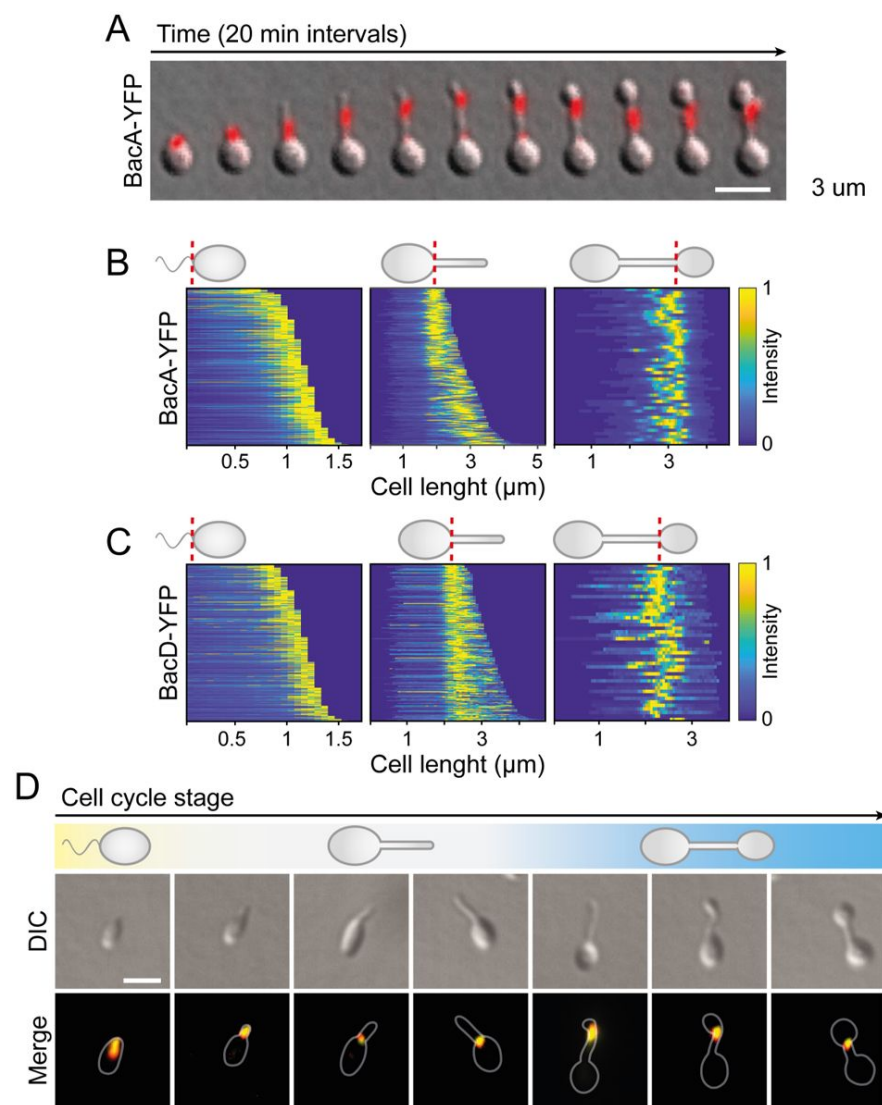


Figure 4.

BacA and BacD show a dynamic, cell cycle-dependent localization pattern.

(A) Localization pattern of BacA-YFP. Cells of strain EC61 (*bacA::bacA-eyfp*) were transferred to agarose pads and imaged at 20-min intervals. Shown are overlays of DIC and fluorescence images of a representative cell, with YFP fluorescence shown in red. Scale bar: 3 μ m. **(B)** Demographic analysis of BacA-YFP localization in swimmer (left), stalked (middle), and budding (right) cells of strain EC61 (*bacA::bacA-eyfp*). The fluorescence intensity profiles measured for cells of each type were sorted according to cell length and stacked on each other, with the shortest cell shown at the top and the longest cell shown at the bottom ($n=250$ swimmer cells, 215 stalked cells and 49 budding cells). Dotted red lines indicate the positions to which cells were aligned. **(C)** Demographic analysis of BacD-Venus localization. Cells of strain EC67 (*bacD::bacD-venus*) were analyzed as described in panel B ($n=416$ swimmer cells, 248 stalked cells and 45 budding cells). **(D)** Co-localization of BacA

and BacD in cells of strain EC68 (*bacA::bacA-eyfp bacD::bacD-mCherry*). Shown are DIC images and overlays of the YFP and mCherry signals of representative cells, arranged according to their developmental state. The individual signals are shown in **Figure S6A**. The Pearson's Correlation Coefficient (PCC) of the two fluorescence signals in a random population of cells ($n=454$) is 0.506. Bar: 1 μ m.

The assembly state of BacA changes at different stages of *H. neptunium* development

Since bactofilins form highly stable polymers *in vitro* (Kühn et al., 2010; Zuckerman et al., 2015), the dynamic localization observed for BacA and BacD was unexpected. To further characterize the dynamics of the bactofilin cytoskeleton in *H. neptunium*, we followed the movement of individual BacA-YFP molecules in swimmer, stalked and budding cells. First, we used single-molecule tracking data to generate high-resolution images of the bactofilin assemblies in each of the three cell types. The results confirmed the cell cycle-dependent localization patterns observed by widefield fluorescence microscopy (Figure 5A). In addition, they revealed clusters in medial regions of the stalk, which may represent transient

polymers formed during the reshuffling of bactofilin molecules between the stalk base and the (incipient) bud neck. When calculating the average mean squared displacement of the tracked BacA-YFP molecules, we found that their mobility decreased gradually from the swimmer over the stalked cell to the budding cell stage (Figure 5B). For each cell type, the distribution of step sizes in the single-particle tracks suggests the existence of two distinct diffusion regimes, with a static ($D = 0.02 \pm 0.0004 \mu\text{m}^2 \text{s}^{-1}$) and a mobile ($D = 0.35 \pm 0.004 \mu\text{m}^2 \text{s}^{-1}$) population, likely representing the polymerized and freely diffusible states, respectively (Figures 5C and Figure 5—figure supplement 1). In swimmer cells, the static fraction comprised only ~60% of the molecules. Its proportion increased to ~70% in stalked cells and finally reached ~80% in budding cells. The F130R variant, by contrast, showed less than 15% of static molecules, consistent with the notion that it is impaired in polymerization but still able to assemble under certain conditions after undergoing a structural change that generates an alternative polymerization interface (Deng et al., 2019). Collectively, these results show that, in *H. neptunium*, the assembly state of the bactofilin cytoskeleton changes as cells progress through their developmental cycle. Moreover, despite the inherent stability of bactofilin polymers, all cell types display a sizeable fraction of mobile BacA-YFP molecules, which could potentially reflect the dynamic reorganization of bactofilin assemblies during cell growth.

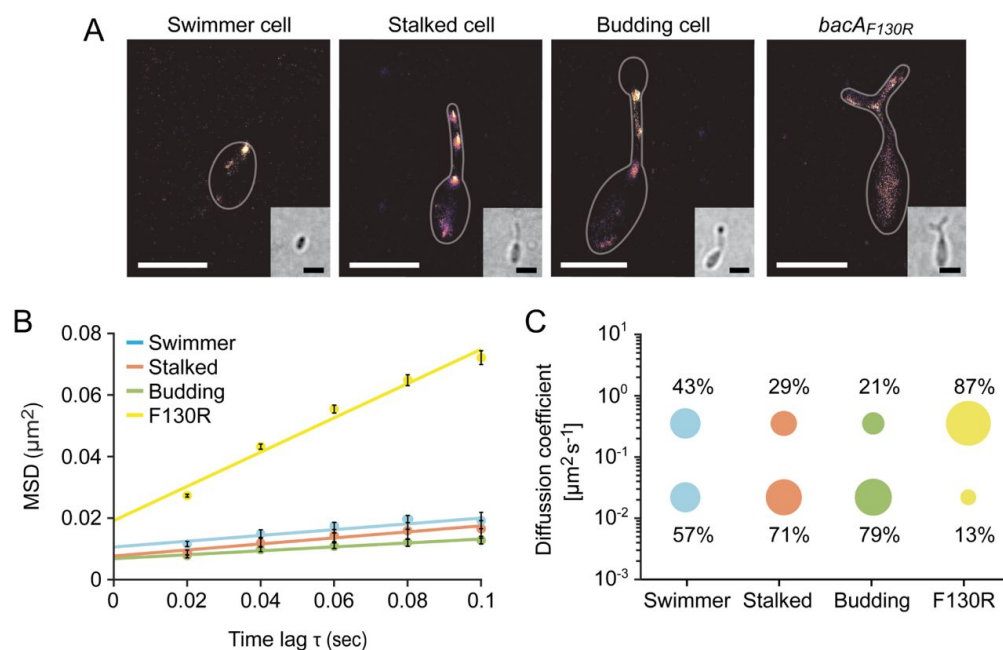


Figure 5.

The single-particle dynamics of BacA change over the course of the cell cycle.

(A) Representative heat maps showing the sum of single-particle positions observed in live-cell single-particle localization microscopy studies of a swimmer, stalked and budding cell producing BacA-YFP (EC61) or a mutant cell producing the polymerization-deficient BacA_{F130R}-YFP variant (MO78). Insets show the

corresponding bright-field images. Bars: 1 μm . (B) Mean-squared displacement (MSD) analysis of the mobility of BacA-YFP in wild-type swimmer ($n=100$), stalked ($n=105$) and budding ($n=101$) cells (EC61). Cells ($n=110$) producing the BacA_{F130R}-YFP were analyzed as a control (MO78). Error bars indicate the standard deviation. (C) Bubble plots showing the proportions of mobile and immobile particles (size of the bubbles) and the corresponding average diffusion constants (y-axis) in the cells analyzed in panel B.

Bactofilin genes are often clustered with genes encoding M23 peptidases

After identifying a critical role of BacA in *H. neptunium* morphogenesis, we set out to gain further insight into its mechanism of action. Notably, in several species in which bactofilins

play a role in cell shape determination, the corresponding bactofilin genes are located adjacent to genes encoding a putative M23 peptidase homologous to LmdC (**Figure 6A**). To determine whether this gene arrangement was more widely conserved, we performed a comprehensive bioinformatic analysis in which we searched all bacterial genome sequences available for putative bactofilin genes that were located immediately upstream or downstream of open reading frames encoding proteins with a predicted M23 peptidase domain. This analysis identified a total of 226 species from a wide variety of bacterial phyla (**Figure 6B** and **Dataset 2**), suggesting a conserved functional association between bactofilins and M23 peptidases.

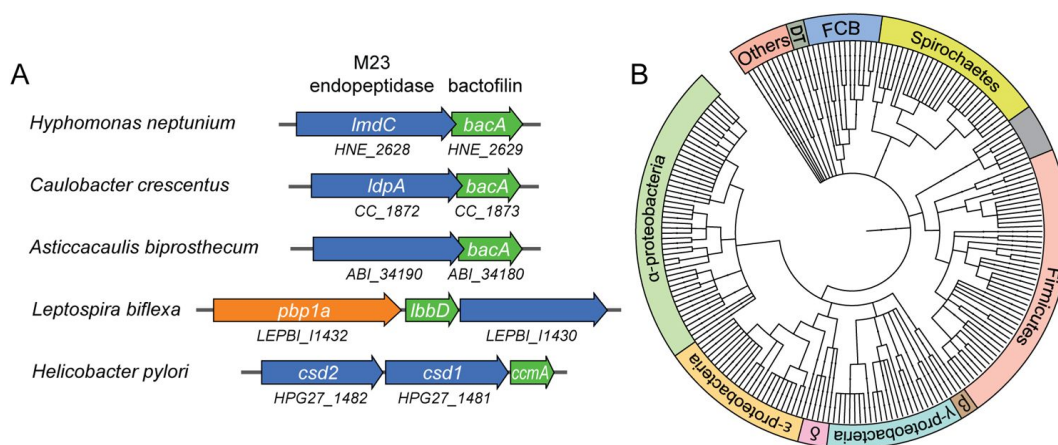


Figure 6.

Putative operons comprising adjacent BacA and M23 peptidase genes are widely conserved among bacteria.

(A) Arrangement of *bacA* and *lmdC* genes in species whose BacA homolog has a proven role in cell morphogenesis. (B) Co-conservation of BacA and M23 peptidase genes across different bacterial phyla. Bioinformatic analysis was used to identify genomes that contained a bactofilin gene located next to an M23 peptidase gene. After retrieval of the taxonomy IDs of the corresponding species from the National Center for Biotechnology Information (NCBI) website, a phylogenetic tree of the species was created using the iTOL server (Letunic and Bork, 2021). Abbreviations: DT (Deinococcus-Thermus group), FCB (Fibrobacteres-Chlorobi-Bacteroidetes group), β (β-proteobacteria), δ (δ-proteobacteria). The full list of species used and details on the genes identified are provided in Dataset 2.

The M23 peptidase LmdC of *H. neptunium* has peptidoglycan hydrolase activity

To clarify whether LmdC was also required for proper growth in *H. neptunium*, we aimed to generate a mutant strain lacking LmdC activity. However, all attempts to delete the *lmdC* gene or the entire putative *lmdC-bacA* operon or to generate non-functional truncated *lmdC* alleles were unsuccessful, in line with a previous report suggesting that *lmdC* is essential for viability (Cserti et al., 2017). It was also not possible to generate a conditional *lmdC* mutant producing the gene under the control of a copper-inducible promoter, suggesting that its expression level needs to be precisely regulated.

Given that LmdC apparently had a critical role in *H. neptunium* growth, we went on to investigate the physiological role of this protein. LmdC is a predicted bitopic membrane protein with a short N-terminal cytoplasmic region, a transmembrane helix, large periplasmic region composed of a predicted coiled-coil domain, and a C-terminal M23 peptidase domain (Figure 7A). Members of the M23 peptidase family usually have Zn²⁺-dependent hydrolase activity and cleave bonds within the peptide side chains of in the peptidoglycan meshwork (Firczuk et al., 2005; Grabowska et al., 2015). However, there are also various representatives that have lost their enzymatic activity because of mutations in residues required for metal cofactor binding and have instead adopted regulatory roles in cell wall biosynthesis (Figueroa-Cuilan et al., 2021; Goley et al., 2010; Gurnani Serrano et al., 2021; Möll et al., 2010; Poggio et al., 2010; Uehara et al., 2010). An amino acid alignment showed that the M23 peptidase domain of LmdC features all residues reported to be critical for Zn²⁺ binding, suggesting that it could act as a genuine peptidoglycan hydrolase (Figure 7B). In a structural model of LmdC generated with AlphaFold2 (Jumper et al., 2021), the periplasmic coiled-coil and M23 peptidase domains form an elongated, rigid unit with two flanking non-structured regions that is flexibly linked to the transmembrane helix (Figure 7C). Notably, LmdC homologs were shown to form a distinct clade among the M23 peptidases that is broadly conserved among species but particularly enriched in alpha- and deltaproteobacteria (Figueroa-Cuilan et al., 2021).

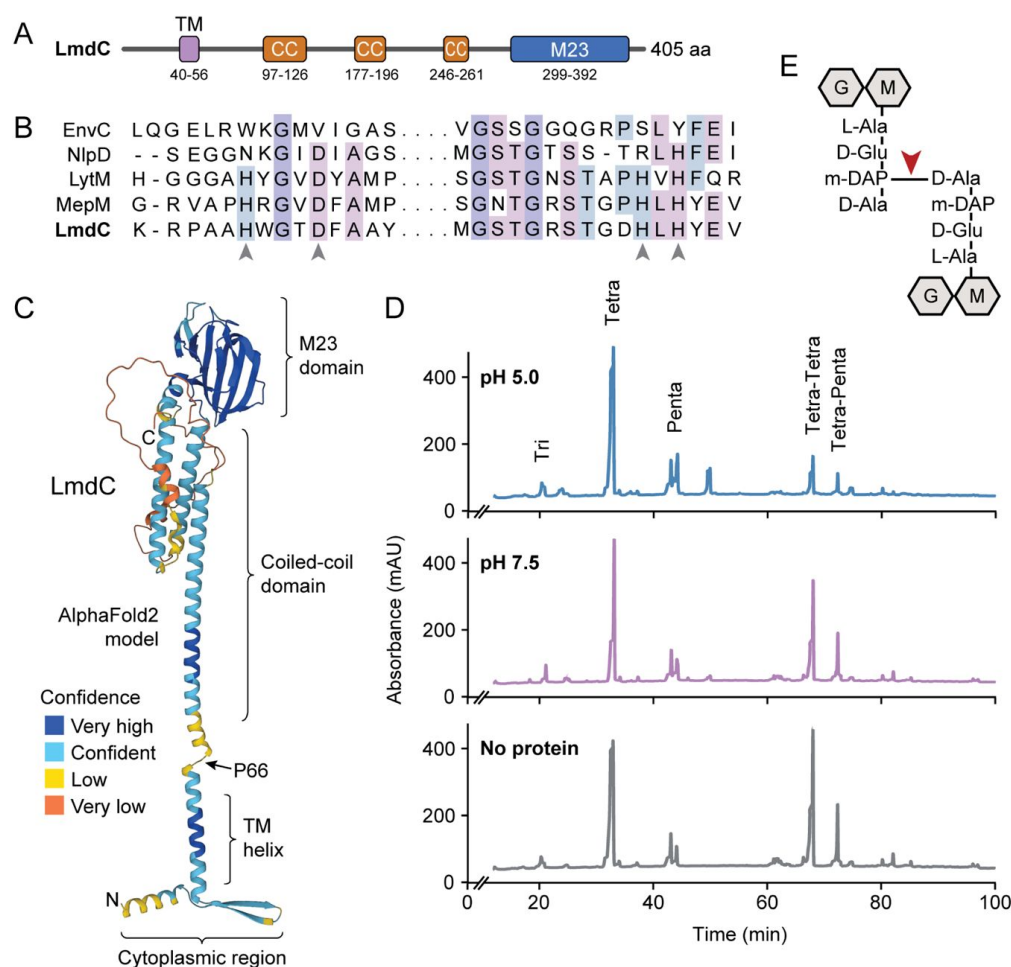


Figure 7.

LmdC is a peptidoglycan hydrolase with DD-endopeptidase activity.

(A) Predicted domain architecture of *H. neptunium* LmdC. The predicted positions of the transmembrane helix (TM), the three coiled-coil regions (CC) and the M23 peptidase domain are indicated. **(B)** Alignment of the amino acid sequences of multiple M23 peptidases showing the conservation of the catalytic residues in LmdC. Residues required to coordinate the catalytic Zn^{2+} ion of LytM from *S. aureus* (Firczuk et al., 2005) are indicated by arrow-heads. The proteins shown are EnvC from *E. coli* (UniProt: P37690), NlpD from *E. coli* (P0ADA3), LytM from *S. aureus* (O33599), MepM from *E. coli* (P0AF59) and LmdC from *H. neptunium* (Q0BYX6). **(C)** Predicted molecular structure of *H. neptunium* LmdC, generated with AlphaFold2 (Jumper et al., 2021). The different domains of the protein and the position of proline 66, which terminates the N-terminal fragment of LmdC used for the *in vivo* analyses in this study (LmdC₁₋₆₆) are indicated. **(D)** HPLC traces showing the mucopeptide profile of peptidoglycan treated with LmdC. Cell walls were incubated with the isolated M23 peptidase domain of LmdC at pH 5.0 and pH 7.5. Subsequently, mucopeptides were released with cellosyl, reduced and separated by HPLC. A control sample lacked LmdC (No protein). The nature of the major products is indicated above the peaks. Tri, Tetra and Penta stand for N-acetylglucosamine-N-acetylmuramitol tripeptide, tetrapeptide and pentapeptide, respectively. **(E)** Structure of a TetraTetra mucopeptide. Abbreviations: G (N-acetylglucosamine), M (N-acetylmuramic acid). The cleavage site of LmdC is indicated by a red arrowhead.

To clarify whether LmdC was indeed catalytically active, we purified a C-terminal fragment of the protein including the M23 peptidase domain and assayed its activity *in vitro*. Peptidoglycan from *H. neptunium* shows only a low degree of cross-linkage and hardly any pentapeptides (Cserti et al., 2017). We therefore used normally crosslinked, pentapeptide-enriched peptidoglycan from *E. coli* strain D456 (lacking PBPs 4, 5 and 6) (Edwards and Donachie, 1993) as a substrate to enable a comprehensive analysis of the cleavage specificity of LmdC. Upon treatment with the protein, the proportion of dimeric mucopeptide species strongly decreased, regardless of the length of the crosslinked peptides and without the formation of major additional monomer peaks (Figure 7D). The activity of LmdC was particularly high at a pH value of 5, but the physiological significance of this effect remains to be determined. Collectively, these results demonstrate that LmdC is a DD-endopeptidase

cleaving the bond between the meso-diaminopimelic acid residue of one peptide and D-alanine at position 4 of the other peptide, thereby reducing the degree of cross-linkage within the peptidoglycan layer (Figure 7E).

LmdC associates with the bactofilin cytoskeleton in *H. neptunium*

To further investigate whether the peptidoglycan hydrolase LmdC was functionally linked to bactofilins in *H. neptunium*, we set out to conduct *in vivo* co-localization studies. All attempts to generate fluorescently labeled variants of full-length LmdC failed, because the fluorescent protein tags were cleaved off after synthesis of the fusion proteins. However, we reasoned that in case LmdC was associated with the cytoplasmic bactofilin cytoskeleton, its recruitment might be mediated by its N-terminal cytoplasmic region, which is predicted to fold into a short α -helix followed by a conspicuous β -hairpin (Figure 7C). To colocalize the two proteins, we therefore generated a truncated reporter construct comprising only the N-terminal cytoplasmic part and the transmembrane helix of LmdC fused to the red fluorescent protein mCherry (Shaner et al., 2004) (Figure 8A). When produced in a *bacA-yfp* background, the fusion protein (LmdC^N-mCherry) formed tight foci that perfectly colocalized with BacA-YFP and showed the characteristic cell cycle-dependent localization pattern observed for the bactofilin assemblies (Figures 8B,C). This result suggests that the N-terminal region of LmdC indeed associates with the bactofilin cytoskeleton, most likely through interaction with its key component BacA. To further validate this hypothesis, we analyzed the interaction between LmdC and BacA *in vitro* by biolayer interferometry. To this end, a synthetic peptide comprising the cytoplasmic region of LmdC (amino acids 1-38) was immobilized on a biosensor and probed with increasing concentrations of purified BacA protein. The results showed that BacA interacts with the LmdC peptide with an apparent equilibrium dissociation constant (K_D) of ~15 μ M. By contrast, an unrelated peptide used as a negative control was not bound with appreciable affinity (Figures 8D,E). This result confirms a direct interaction between BacA and LmdC and identifies the N-terminal cytoplasmic region of LmdC as the interaction determinant for BacA association. Notably, however, LmdC^N-mCherry still formed foci in bactofilin-deficient cells, albeit at apparently random positions, suggesting that the protein can assemble into larger complexes independently of its interaction with BacA (Figure 8—figure supplement 1).

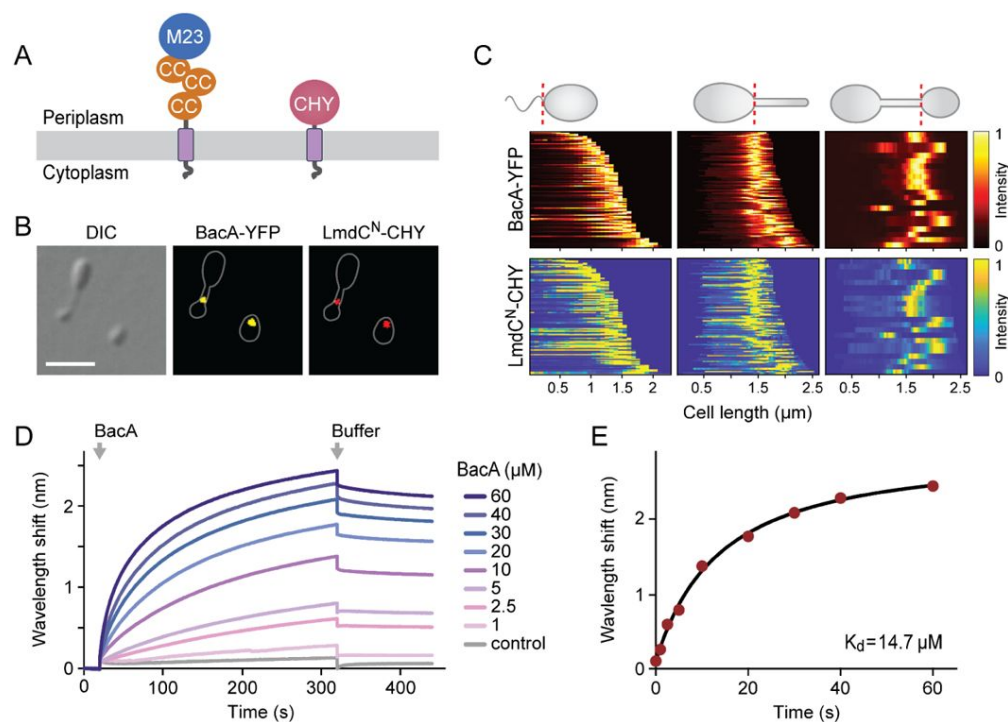


Figure 8.

LmdC colocalizes and directly interacts with BacA.

(A) Domain architectures of the native LmdC protein and the LmdC₁₋₆₆-mCherry fusion used for co-localization studies. The transmembrane helix is shown in purple. Abbreviations: CC (coiled-coil domain), M23 (M23 peptidase domain), CHY (mCherry). (B) Co-localization of BacA-YFP and LmdC^N-mCherry in *H. neptunium*. Shown are representative cells of strain MO79 (*bacA::bacA-eyfp* P_{Cu}::P_{Cu}-*lmdC^N-mCherry*) that were in-

duced for 2 h with 0.5 mM CuSO₄ prior to analysis by DIC and fluorescence microscopy. (C) Demographic analysis of BacA-YFP and LmdC^N-mCherry co-localization in swimmer (left; n=99), stalked (middle; n=103) and budding (right; n=22) cells from the culture imaged in panel B. Data were analyzed as described for Figure 4B. Corresponding fluorescence profiles are shown at the same relative position in the two graphs. (D) Bio-layer interferometric analysis of the interaction between cytoplasmic domain of LmdC and BacA. Sensors derivatized with a biotinylated synthetic peptide comprising the N-terminal cytoplasmic region of LmdC (amino acids 1-38) were probed with the indicated concentrations of BacA. After the association of BacA, the sensors were transferred to protein-free buffer to induce BacA dissociation. The interaction kinetics were followed by monitoring the wavelength shifts resulting from changes in the optical thickness of the sensor surface during association or dissociation of the analyte. The extent of non-specific binding of BacA to the sensor surface was negligible (control). (E) Affinity of the BacA-LmdC interaction. The maximal wavelength shifts measured in the experiment shown in panel D were plotted against the corresponding BacA concentration. The data were fitted to a one-site binding model, yielding an apparent K_d value of ~15 μ M.

R. rubrum BacA recruits LmdC to the inner cell curvature to modulate spiral cell shape

Our bioinformatic analysis suggests that bactofilins and M23 peptidases may be functionally associated in a large number of species. To further explore this possibility, we turned our efforts to the spiral-shaped bacterium *Rhodospirillum rubrum*, which contains a putative *lmdC-bacA* operon similar to that in *H. neptunium* (Munk et al., 2011) (Figure 9A). The *R. rubrum* BacA homolog Rru_A1867 (BacA_{RS}) is 37% identical (57% similar) to BacA of *H. neptunium* and also consists of a central Bactofilin A/B domain flanked by terminal non-structured regions (Figure 9B). The LmdC homolog Rru_A1868 (LmdC_{RS}) is 39% identical (55% similar) to *H. neptunium* LmdC and has a similar predicted molecular structure (Varadi et al., 2022). The deletion of *bacA_{RS}* led to a noticeable increase in cell curvature (Figures 9C and Figure 9—figure supplement 1A), as also reflected in a significant increase in cell sinuosity (Figure 9D). A very similar effect was observed for cells lacking *lmdC_{RS}* or both

bacA_{RS} and *lmdC_{RS}*, indicating that the two gene products act in the same pathway and jointly contribute to cell shape maintenance in *R. rubrum* (Figures 9C,D and Figure 1—figure supplement 1A). In the two mutant strains, the global composition of peptidoglycan was largely unchanged, supporting the notion that the *BacA_{RS}*-*LmdC_{RS}* pathway modifies the cell wall only locally or acts by modulating the activity of the generic cell elongation machinery (Figure 9—figure supplement 1B).

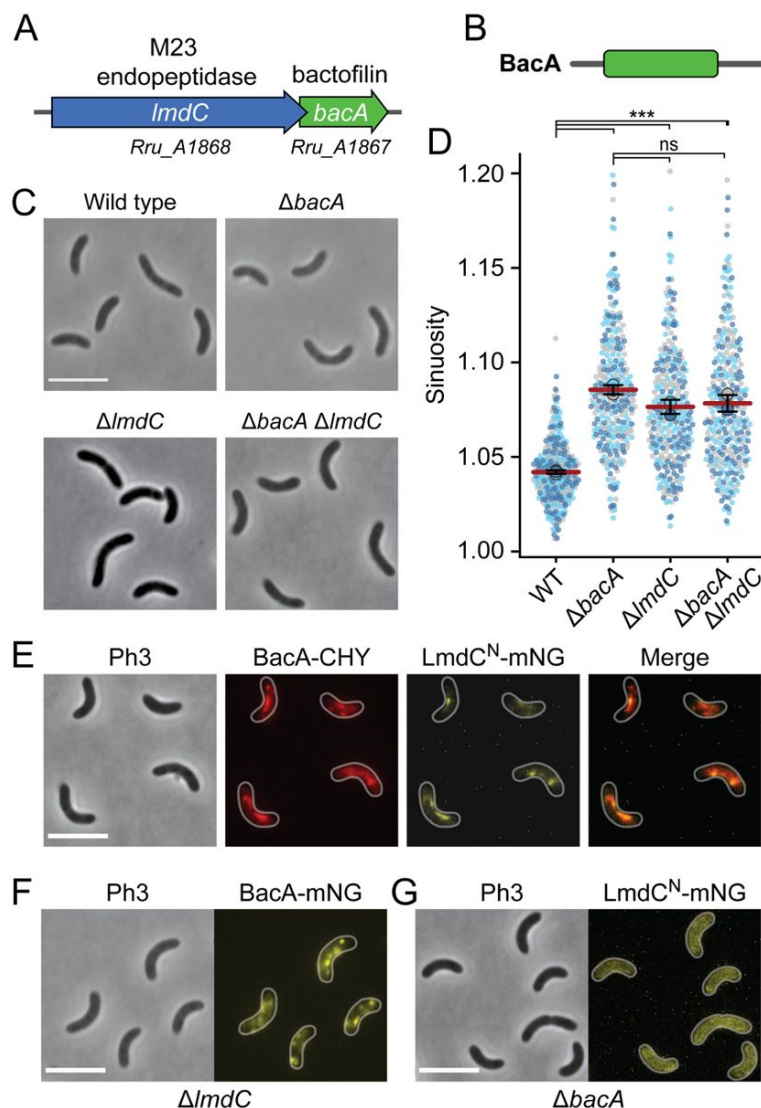


Figure 9.

BacA and LmdC contribute to cell morphogenesis in *R. rubrum*.

(A) Schematic representation of the *bacA-lmdC* operon in *R. rubrum*. (B) Domain organization of *BacA* from *R. rubrum*. The bactofilin polymerization domain (green box) is flanked by non-structured N- and C-terminal regions. (C) Phenotypes of *R. rubrum* wild-type (S1), $\Delta bacA_{RS}$ (SP70), $\Delta lmdC_{RS}$ (SP68) and $\Delta bacA_{RS} \Delta lmdC_{RS}$ (SP116) cells, imaged using phase-contrast microscopy. Bar: 5 μm . (D) Superplots showing the distribution of cell sinuosity in populations of the indicated *R. rubrum* strains. Small dots represent the data points, large dots represent the median values of the three independent experiments shown in the graphs (dark blue, light blue, grey). The mean of the three median values is indicated by a horizontal red line. Whiskers represent the standard deviation. *** $p < 0.005$; ns, not significant; t-test. (E) Co-localization of *BacA_{RS}*-mCherry and *LmdC_{RS}*^N-mNeonGreen in the $\Delta lmdC_{RS}$ background (SP119). The Pearson's Correlation Coefficient (PCC) of the two fluorescence signals in a random subpopulation of cells ($n=87$) is 0.515. Scale bar: 5 μm . (F) Localization of *BacA_{RS}*-mNeonGreen in the $\Delta lmdC_{RS}$ background (SP98). Bar: 5 μm . (G) Localization of *LmdC_{RS}*-mNeonGreen in the $\Delta bacA_{RS}$ background (SP118). Bar: 5 μm .

Complementation studies showed that the expression of a plasmid-borne *bacA* copy under the control of a weak constitutive promoter attenuated the curvature defect, albeit only partially, likely due to inadequate expression levels (Figure 9—figure supplement 2). The generation of an analogous plasmid to complement the deletion of *lmdC_{RS}* failed, because it was not possible to introduce the construct into *Escherichia coli* for plasmid propagation, suggesting that the expression of *lmdC_{RS}* was toxic to the cells. To further test this hypothesis, we constructed a plasmid enabling the inducible expression of the gene in *E. coli*. Upon induction, a large part of the cell population lysed, whereas control cells carrying an empty plasmid did not show any phenotypic defects (Figure 9—figure supplement 3). These results indicate that, similar to its *H. neptunium* homolog, *LmdC_{RS}* is an active enzyme with peptidoglycan hydrolase activity.

To determine how $BacA_{RS}$ and $LmdC_{RS}$ jointly affect cell curvature, we first generated an *R. rubrum* strain that lacked $lmdC_{RS}$ and carried a fully functional $bacA_{RS}$ -mCherry fusion (Figure 9—figure supplement 2) in place of the native $bacA_{RS}$ gene. Subsequently, we introduced a replicating plasmid that harbored a constitutively expressed reporter construct encoding the predicted N-terminal cytoplasmic region and the transmembrane helix (amino acids 1-80) of $LmdC_{RS}$ fused to the green fluorescent protein mNeonGreen (Shaner et al., 2013) ($LmdC_{RS}^N$ -mNG) (Figure 9—figure supplement 4A,B). We found that the two proteins colocalized, forming patchy or filamentous structures that were preferentially, but not exclusively, positioned at the inner cell curvature (Figure 9E). A very similar pattern was observed in the presence of the wild-type $lmdC_{RS}$ gene (Figure 9—figure supplement 4C). Importantly, the localization pattern of $BacA_{RS}$ was not affected by the absence of $LmdC_{RS}$ (Figure 9F), whereas the $LmdC_{RS}^N$ -mNG fusion was completely dispersed in cells lacking $BacA_{RS}$, indicating that $BacA_{RS}$ acts as a localization determinant for $LmdC_{RS}$ (Figure 9G). Collectively, these findings confirm a functional association between the bactofilin cytoskeleton and $LmdC_{RS}$ in *R. rubrum*. Moreover, they lend support to the idea that bactofilins and M23 peptidases form a conserved module involved in bacterial cell shape determination.

Discussion

Although bactofilins are widely conserved among bacteria, their cellular functions are still incompletely understood. Here, we show that $BacA$ homologs critically contribute to cell shape determination in two morphologically distinct members of the alphaproteobacteria, the stalked budding bacterium *H. neptunium* and the spiral-shaped bacterium *R. rubrum*. In both cases, they functionally interact with $LmdC$ -type DD-endopeptidases to promote local changes in the pattern of peptidoglycan biosynthesis. Bactofilins thus complement the activities of the MreB and FtsZ cytoskeletons in the regulation of cell growth and act as versatile cell shape modifiers that facilitate the establishment of complex bacterial morphologies.

In *H. neptunium*, the bactofilin cytoskeleton localizes dynamically to the stalk base and the bud neck, promoting stalk growth and subsequent bud formation (Figure 4). While flanked by zones of active growth, the stalk itself is usually devoid of cell wall biosynthetic machinery (Cserti et al., 2017). In bactofilin-deficient cells, however, newly formed stalks are remodeled into wide and often branched cellular extensions that grow in an apparently uncontrolled manner (Figure 2 and Figure 2—figure supplement 1). This observation suggests that bactofilin polymers establish barriers that normally prevent the entry of elongasome complexes from the mother cell or nascent bud compartments into the stalk. In their absence, the stalk and bud growth zones are no longer confined and expand into the stalk envelope, leading to stalk widening and the formation of irregular bulges due to uncontrolled peptidoglycan incorporation and bud expansion. Bulges or larger amorphous segments at some point separate from the mother cell and then continue to grow as independent cells, indicating that chromosome replication and segregation as well as cell division still occur under these conditions.

The precise mechanism underlying the function of bactofilin in cell morphogenesis remains to be clarified. It is conceivable that bactofilin polymers act by tethering peptidoglycan biosynthetic proteins or establishing physical barriers that hinder the mobility of elongasome complexes (Figure 10A). However, we observed a close functional association of bactofilins with M23 peptidases that appears to be widely conserved among species, suggesting that their activity may be intimately tied to cell wall hydrolysis. The transition zones between the stalk and the adjacent mother cell and bud compartments are

characterized by a high degree of positive cell envelope curvature, which is in stark contrast to the negative curvature in the remaining parts of the cell. The hydrolytic activity of *H. neptunium* LmdC may be critical to remodel the cell wall in these zones. Importantly, MreB is thought to move along regions of negative inner curvature (Hussain et al., 2018; Wong et al., 2019). The positively curved transition zones generated by the bactofilin-LmdC assemblies could therefore represent topological barriers that are difficult to cross by elongasome complexes, thereby helping to restrict their movement to the mother cell and bud compartments. This effect may also be important during the stalk elongation phases following cell division, when bactofilin assemblies are no longer present at the stalk base.

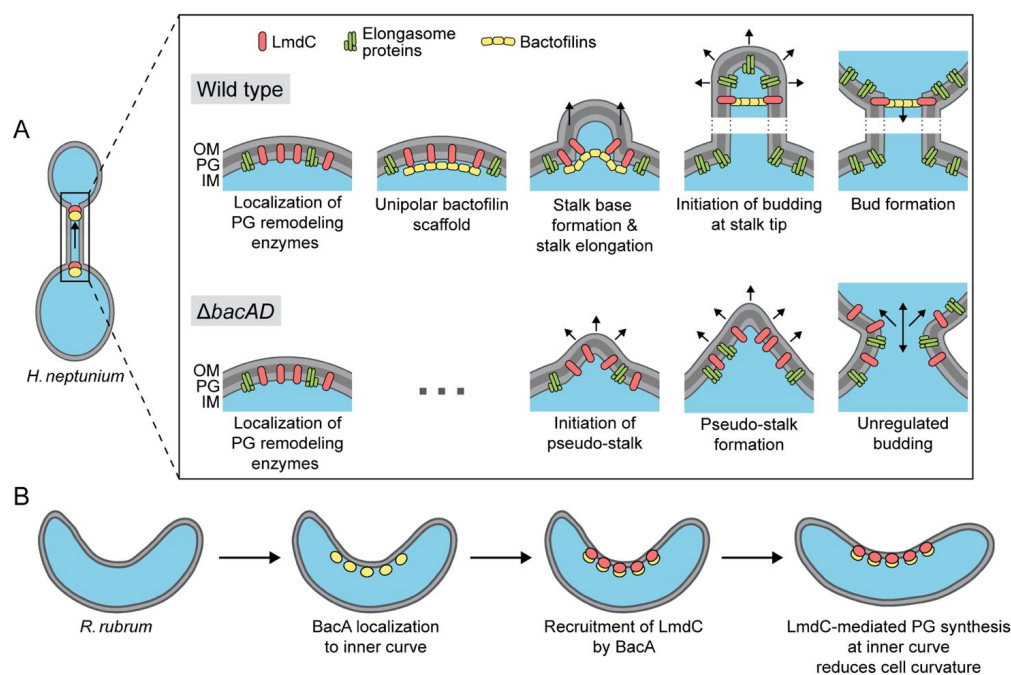


Figure 10.

Model of the roles of BacA and LmdC in *H. neptunium* and *R. rubrum* cell morphogenesis.

(A) Cell morphogenesis in *H. neptunium* wild-type (top) and $\Delta bacAD$ cells (bottom). In wild-type cells, BacA polymers form a complex with the DD-endopeptidase LmdC at the future stalked pole. The hydrolytic activity of LmdC helps to curve the cell wall at the incipient stalk base and thus determine stalk morphology. The

zone of high positive inner cell curvature, and potentially the physical barrier constituted by the bactofilin polymer, prevent the movement of elongasome complexes from the mother cell body into the stalk, thereby limiting peptidoglycan biosynthesis to the stalked cell pole. At a later stage, the bactofilin-LmdC complex localizes close to the tip of the stalk and again generates a ring-shaped zone of positive cell curvature, thereby promoting the remodeling of the stalk tip into a spherical bud. At the onset of the budding process, elongasome complexes accumulate in the nascent bud by a so-far unknown mechanism. The positively curved bud neck and the bactofilin polymers prevent the movement of elongasome complexes from the bud into the stalk, thereby limiting cell growth to the bud compartment. In the $\Delta bacAD$ mutant, cells fail to concentrate LmdC at the future stalk base and bud neck. As a consequence, peptidoglycan biosynthesis is no longer limited to the different growth zones, leading to pseudo-stalk formation and unregulated bud expansion. (B) Bactofilin-mediated modulation of cell helicity in *R. rubrum*. BacA_{RS} (yellow) recruits LmdC_{RS} (red) to the inner cell curvature. The hydrolytic activity of LmdC ultimately stimulates peptidoglycan biosynthesis at this position, leading to a reduction in cell helicity.

Notably, a role in stalk formation has also been reported for the bactofilin homologs of *C. crescentus* and *A. biprosthecum*. In *C. crescentus*, the major bactofilin BacA was shown to act as a localization determinant for the bifunctional penicillin-binding protein PbpC, a cell wall synthase that contributes to stalk elongation (Kühn et al., 2010) and the proper sorting of a stalk-specific protein (Hughes et al., 2013). However, the absence of BacA only leads to a moderate reduction in stalk length and leaves overall cell shape unaffected, indicating that this protein has only a minor role in *C. crescentus* stalk formation. In *A. biprosthecum*, by

contrast, the deletion of *bacA* completely abolishes stalk formation in rich medium and leads to the formation of wide cellular protrusions, named pseudostalks, under phosphate-limiting conditions (Caccamo et al., 2020). Similar to the amorphous extensions observed for an *H. neptunium* $\Delta bacAD$ mutant, these structures grow through disperse incorporation of new peptidoglycan and develop into viable offspring. Their formation was also attributed to the entry of cell wall biosynthetic proteins into nascent stalks (Caccamo et al., 2020), suggesting that the *H. neptunium* and *A. biprosthicum* BacA homologs share the same barrier function during stalk formation. However, in *H. neptunium*, another layer of regulation has been added in which bactofilins establish a second barrier close to the stalk tip that enables the formation of stalk-terminal buds. Bud expansion requires the relocation of elongasome complexes from the mother cell to the nascent bud at the onset of bud formation, as likely reflected by the fact that components of the elongasome (MreB, RodZ) and new cell wall biosynthesis can be detected within the stalk during a short interval at the end of the stalk elongation phase (Cserti et al., 2017). This process may be facilitated by the (partial) disassembly of the bactofilin complex at the stalk base, but the mechanistic details remain to be investigated. Another open question concerns the factors that control the dynamic localization of bactofilin to the stalked pole and the future bud neck in the terminal segment of the stalk. In *A. biprosthicum*, the recruitment of BacA assemblies to the sites of stalk biosynthesis is dependent on the cell wall hydrolase SpmX (Caccamo et al., 2020). However, an *H. neptunium* mutant lacking this protein shows normal morphology (Leicht et al., 2020), suggesting the existence of a different localization determinant. The contribution of LmdC to bactofilin localization could not be investigated, because this protein is essential for *H. neptunium* growth (Cserti et al., 2017). However, the apparently random positioning of LmdC complexes in bactofilin-deficient cells suggests that its recruitment depends on the bactofilin cytoskeleton, excluding it as a potential candidate.

Intriguingly, bactofilins not only mediate stalk formation and stalk-terminal budding but also the formation of curved cell shapes. We show that both BacA and LmdC are required to establish the normal degree of cell helicity in *R. rubrum*, with their absence leading to hyper-curved cells (Figure 9C,D). This result suggests that the two proteins form an accessory module that counteracts the activity of a so-far unknown system responsible for generating spiral cell shape in *R. rubrum* (Figure 10B). Cell curvature was shown to promote cell motility (Martinez et al., 2016). It is tempting to speculate that the expression of *bacA* and *lmdC* could be controlled in response to external cues to ensure optimal cell helicity and, thus, fitness in different environments. Our results show that *R. rubrum* cells contain multiple BacA-LmdC complexes that are distributed along the entire cell envelope but localize preferentially to the inner cell curvature (Figure 9E). In this case, it is immediately evident that BacA acts as a localization determinant for LmdC, since LmdC is evenly dispersed in its absence (Figure 9G). To reduce cell helicity, the hydrolytic activity of LmdC must ultimately stimulate the insertion of new peptidoglycan at the inner curvature, thereby increasing the rate of cell elongation in this region and straightening the cell. A similar mechanism may also be at work in the spirochete *L. biflexa*, where removal of the bactofilin paralog LbbD, whose gene lies adjacent to a gene for putative M23 peptidase (LEPBI_I1430) (Figure 3A), was found to induce a strong increase in cell helicity (Jackson et al., 2018). A different variation of this theme is found in *H. pylori*. There, the bactofilin homolog CcmA also forms multiple assemblies along the cell envelope, which interact with a membrane-spanning protein complex including an M23 peptidase (Csd1) homologous to LmdC (Sichel et al., 2022; Taylor et al., 2020). However, these assemblies are enriched at the outer curvature, where they stimulate peptidoglycan synthesis to locally increase the rate of cell elongation over that at the inner curvature, leading to twisting of the cell body.

Notably, in all systems characterized so far in molecular detail, the function of bactofilin-M23 peptidase complexes involves a stimulation of cell wall biosynthesis in a confined region of the cell envelope that entails a local change in cell envelope curvature. In *H. neptunium*, these complexes are only localized to narrow bands at the stalk base and bud

neck, generating sharp bends at the transition zones between the stalk and the adjacent mother cell and bud compartments. In spiral-shaped bacteria, by contrast, they are scattered along the entire length of the cell, thereby establishing an elongated zone of increased longitudinal growth that changes overall cell curvature. The same mechanistic principle may also explain the morphogenetic role of bactofilins in so-far uninvestigated systems.

Collectively, our results underscore the role of bactofilins as versatile modulators of bacterial cell shape. In association with M23 peptidases, they form cell wall biosynthetic complexes that introduce local changes in cell envelope curvature. In doing so, they complement the activities of the tubulin and actin cytoskeletons in cell shape determination and thus expand the morphogenetic potential of bacteria, enabling the generation of complex cell shapes that go beyond the generic rod-like or spherical morphologies. It will be interesting to investigate the molecular function of bactofilins in a range of morphologically diverse species to obtain a comprehensive picture of the functionalities provided by these cytoskeletal proteins and the conservation of their mode of action.

Materials and methods

Media and growth conditions

H. neptunium LE670 (ATCC 15444) (Leifson, 1964) and its derivatives were grown in Artificial Sea Salt (ASS) medium at 28 °C under aerobic conditions, shaking at 210 rpm in baffled flasks. ASS medium was composed of 0.5% Bacto Peptone (Thermo Fisher Scientific, USA), 0.1% yeast extract, 1 mM MgSO₄, 0.5 mM CaCl₂ and 1.5% Instant Ocean Sea Salt (Spectrum Brands, USA), dissolved in deionized water. For solid media, 1.5% agar was added prior to autoclaving. When appropriate, antibiotics were used at the following concentrations (µg/mL in liquid/solid medium): rifampicin (1/2), kanamycin (100/200). The expression of genes under the control of the copper-inducible P_{Cu} promoter or the zinc-inducible P_{Zn} promoter (Jung et al., 2015) was induced by the addition of CuSO₄ or ZnSO₄ to the concentrations indicated in the text. To assess the growth of *H. neptunium*, cells were grown to exponential phase, diluted in fresh medium to an optical density at 580 nm (OD₅₈₀) of 0.02, and transferred into 24-well polystyrene microtiter plates (Becton Dickinson Labware, USA). Growth was then followed at 32 °C under double-orbital shaking in an EPOCH 2 microplate reader (BioTek, USA) by measuring the OD₅₈₀ at 20-min intervals.

R. rubrum S1 (ATCC 11170) (Molisch, 1907; Pfennig and Trüper, 1971; van Niel, 1944) and its derivatives were grown in Bacto Tryptic Soy Broth (BD Diagnostic Systems, USA) at 28 °C under aerobic conditions, shaking at 210 rpm in Erlenmeyer flasks. When appropriate, media were supplemented with antibiotics at the following concentrations (µg/mL in liquid/solid medium): kanamycin (30/30), cefalexin (15/-).

E. coli strains were cultivated aerobically (shaking at 210 rpm) at 37 °C in LB medium. For plasmid-bearing strains, antibiotics were added at the following concentrations (µg/mL in liquid/solid medium): kanamycin (30/50), rifampicin (25/50), ampicillin (50/200). To grow *E. coli* WM3064, media were supplemented with 2,6-diaminopimelic acid (DAP) to a final concentration of 300 µM.

Plasmid and strain construction

The bacterial strains, plasmids, and oligonucleotides used in this study are listed in **Tables S3-S6**. *E. coli* TOP10 (Thermo Fisher Scientific, USA) was used as host for cloning purposes. All plasmids were verified by DNA sequencing. *H. neptunium* and *R. rubrum* were transformed by conjugation using the DAP-auxotrophic strain *E. coli* WM3064 as a donor

(Jung et al., 2015). The integration of non-replicating plasmids at the chromosomal P_{Cu} or P_{Zn} locus of *H. neptunium* was achieved by single homologous recombination (Jung et al., 2015). Gene replacement in *H. neptunium* and *R. rubrum* was achieved by double-homologous recombination using the counter-selectable *sacB* marker (Cserti et al., 2017). Proper chromosomal integration or gene replacement was verified by colony PCR.

Live-cell imaging

Cells of *H. neptunium* and *R. rubrum* were grown to exponential phase in the appropriate medium and, if suitable, induced with CuSO_4 prior to imaging. For depletion experiments, cells were grown in the presence of inducer, washed three times with inducer-free medium and then further cultivated in the absence of inducer for the indicated period of time prior to analysis. To acquire still images, cells were transferred onto 1% agarose pads (in water). For time-lapse analysis, cells were immobilized on pads made of 1% agarose in ASS medium, and the cover slides were then sealed with VLAP (1:1:1 mixture of vaseline, lanolin, and paraffin) to prevent dehydration. Imaging was performed using a Zeiss Axio Imager.M1 microscope equipped with a Zeiss alpha Plan-Apochromat 100x/1.40 Oil DIC objective and a pco.edge 3.1 sCMOS camera (PCO, Germany) or a Zeiss Axio Imager.Z1 microscope equipped with a Zeiss alpha Plan-Apochromat 100x/1.46 Oil DIC M27 or a Plan-Apochromat 100x/1.40 Oil Ph3 M27 objective and a pco.edge 4.2 sCMOS camera (PCO, Germany). An X-Cite 120PC metal halide light source (EXFO, Canada) and ET-DAPI, ET-YFP or ET-TexasRed filter cubes (Chroma, USA) were used for fluorescence detection. Microfluidic experiments were performed using a CellASIC ONIX EV262 Microfluidic System, equipped with an F84 manifold and B04A microfluidic plates (Merck Millipore, Germany), which were flushed with PBS buffer for 30 min before usage. Exponentially growing cells were flushed into the flow cells, cultivated under continuous medium flow (0.4 ml/h), and imaged at regular intervals using the Axio Image.Z1 microscope described above. To visualize sites of ongoing peptidoglycan synthesis, cells were grown to the exponential phase, incubated for 1 min with 1 mM HADA and washed four times in ASS medium prior to imaging.

Images were recorded with VisiView 3.3.0.6 (Visitron Systems, Germany) and processed with ImageJ (Schneider et al., 2012) and Adobe Illustrator CS6 (Adobe Systems, USA). The subcellular distribution of fluorescence signals was analyzed with BacStalk (Hartmann et al., 2020). Pearson's correlation coefficients were determined using the JACoP plug-in (Bolte and Cordelières, 2006) for ImageJ. Cell sinuosity was determined using the ImageJ plug-in MicrobeJ (Ducret et al., 2016), with each analysis performed in triplicate and 100 cells analyzed per strain and experiment. Results were displayed as SuperPlots (Lord et al., 2020) generated using the SuperPlotsOfData web application (Goedhart, 2021). Pearson's correlation coefficients were determined with ImageJ, using the "Align Fluorescence Channels" plugin (Norbert Vischer, unpublished) to align the fluorescence images to be compared and then the JACoP plugin (Fabrice P. Cordelières, unpublished) to determine the degree of colocalization.

Single-particle tracking and diffusion analysis

Cells were cultivated overnight in ASS medium, transferred into fresh medium and grown to exponential phase prior to imaging by slimfield microscopy (Plank et al., 2009). In this approach, the back aperture of the objective is underfilled by illumination with a collimated laser beam of reduced width, generating an area of $\sim 10 \mu\text{m}$ in diameter with a light intensity high enough to enable the visualization of single fluorescent protein molecules at very high acquisition rates. The single-molecule level was reached by bleaching of most molecules in the cell for 100 to 1,000 frames, followed by tracking of the remaining and newly synthesized molecules for $\sim 3,000$ frames. Images were taken at 30 ms intervals using an Olympus IX-71 microscope equipped with a UAPON 100x/ NA 1.49 TIRF objective, a back-illuminated

electron-multiplying charge-coupled device (EMCCD) iXon Ultra camera (Andor Solis, USA) in stream acquisition mode, and a LuxX 457-100 (457 nm, 100 mW) light-emitting diode laser (Omicron-Laserage Laserprodukte GmbH, Germany) as an excitation light source. The laser beam was focused onto the back focal plane and operated during image acquisition with up to 2 mW (60 W/cm² at the image plane). Andor Solis 4.21 software was used for camera control and stream acquisition. Prior to analysis, frames recorded before reaching the single-molecule level were removed from the streams, using photobleaching curves as a reference. Subsequently, the streams were cropped to an equal length of 2,000 frames and the proper pixel size (100 nm) and time increment were set in the imaging metadata using Fiji (Schindelin et al., 2012). Single particles were tracked with u-track 2.2 (Jaqaman et al., 2008). Trajectories were only considered for further statistical analysis if they had a length of at least five steps. Data analysis was performed using SMTracker 2.0 (Oviedo-Bocanegra et al., 2021). An estimate of the diffusion coefficient and insight into the kind of diffusive motion exhibited were obtained from mean-squared-displacement (MSD)-versus-time-lag curves. In addition, the frame-to-frame displacements of all molecules in x and the y direction were fitted to a two-population Gaussian mixture model to determine the proportions of mobile and static molecules in each condition (Oviedo-Bocanegra et al., 2021).

Transmission electron microscopy

Images of *H. neptunium* cells, 10 µl of early exponential cell cultures were applied onto glow-discharged electron microscopy grids (Formvar/Carbon Film on 300 Mesh Copper; Plano GmbH, Germany) and incubated for 1 min at room temperature. The grid was manually blotted with Whatman filter paper to remove excess liquid. Subsequently, the cells were negatively stained for 5 sec with 5 µl of 1% uranyl acetate. After three washes with H₂O, the grids were dried and imaged in a 100 kV JEM-1400 Plus transmission electron microscope (JEOL, USA). To image BacA polymers, purified BacA-His₆ was dialyzed against low-salt buffer (50 mM HEPES pH 7.2, 10 mM NaCl, 5 mM MgCl₂, 0.1 mM EDTA, 1 mM β-mercaptoethanol) for 16 h. The protein was spotted onto carbon-coated grids and allowed to settle for 2 min. The grids were blotted dry, stained with 1:2 diluted supernatant of saturated 2 % uranyl acetate (in H₂O) for 1 min, dried and imaged using a Zeiss CEM902 electron microscope, operated at 80 kV and equipped with a 2048×2048 pixel CCD camera. Image processing was carried out using Photoshop CS2 and Illustrator CS5 (Adobe Systems, USA).

Immunoblot analysis

Antibodies against BacA were raised by immunization of rabbits with purified BacA-His₆ protein (Eurogentec, Belgium). Cells were harvested in the exponential growth phase. Immunoblot analysis was conducted as described previously (Thanbichler and Shapiro, 2006), using anti-BacA antiserum (1:10,000), a monoclonal anti-mNeonGreen antibody (Chromotek, Germany; Cat. #: 32f6; RRID: AB_2827566), a polyclonal anti-GFP antibody (Sigma, Germany; Cat. #: G1544; RRID: AB_439690) or a polyclonal anti-mCherry antibody (BioVision, USA; Cat. #: 5993; RRID: AB_1975001) at dilutions of 1:10,000, 1:1,000, 1:10,000 or 1:10,000, respectively. Goat anti-rabbit immunoglobulin G conjugated with horse-radish peroxidase (Perkin Elmer, USA) was used as secondary antibody. Immunocomplexes were detected with the Western Lightning Plus-ECL chemiluminescence reagent (Perkin Elmer, USA). The signals were recorded with a ChemiDoc MP imaging system (BioRad, Germany) and analyzed using Image Lab software (BioRad, Germany).

Peptidoglycan analysis

Cultures of exponentially growing cells of the *H. neptunium* wild type and its mutant derivatives EC28 (Δ*bacA*), EC23 (Δ*bacD*) and EC33 (Δ*bacAD*) were rapidly cooled to 4°C and

harvested by centrifugation at 16,000 ×g for 30 min. The cells were resuspended in 6 ml of ice-cold H₂O and added dropwise to 6 ml of a boiling solution of 8% sodium dodecylsulfate (SDS) that was stirred vigorously. After 30 min of boiling, the suspension was cooled to room temperature. Peptidoglycan was isolated from the cell lysates as described previously (Glauner, 1988) and digested with the muramidase cellosyl (kindly provided by Hoechst, Frankfurt, Germany). The resulting muropeptides were reduced with sodium borohydride and separated by HPLC following an established protocol (Bui et al., 2009; Glauner, 1988). The identity of eluted fragments was assigned based on the known retention times of muropeptides, as reported previously (Cserti et al., 2017; Glauner, 1988).

LmdC enzymatic activity assay

To test the enzymatic activity of LmdC, peptidoglycan from *E. coli* D456 (Edwards and Donachie, 1993) was mixed with 10 μM LmdC in buffer A (20 mM Hepes/NaOH pH 7.5, 50 mM NaCl, 1 mM ZnCl₂) or buffer B (20 mM sodium acetate pH 5.0, 50 mM NaCl, 1 mM ZnCl₂) in a final volume of 50 μL and incubated at 37°C for 16 h in a thermal shaker set at 900 rpm. A mixture of peptidoglycan in buffer B without LmdC was used as a control. The reactions were stopped by heating at 100°C for 10 min. The peptidoglycan was then further digested overnight with cellosyl, and the reactions were stopped by heating at 100°C for 10 min. After centrifugation of the samples at 14,000 rpm for 10 min, the supernatants were recovered, reduced with sodium borohydride, acidified to pH 4.0 – pH 4.5 with dilute 20% phosphoric acid and subjected to HPLC analysis as previously described (Glauner, 1988).

Bio-layer interferometry

Bio-layer interferometry analyses were conducted using a BLItz system equipped with High Precision Streptavidin (SAX) Biosensors (Sartorius, Germany). As a ligand, a custom-synthesized N-terminally biotinylated peptide comprising residues Met1 to Gln38 of LmdC (GenScript, USA) was immobilized on the biosensors. After the establishment of a stable baseline, association reactions were monitored at various analyte concentrations. At the end of each binding step, the sensor was transferred into an analyte-free buffer to follow the dissociation kinetics. The extent of non-specific binding was assessed by monitoring the interaction of analyte with unmodified sensors. All analyses were performed in BLItz binding buffer (25 mM HEPES/KOH pH 7.6, 100 mM KCl, 10 mM MgSO₄, 1 mM DTT, 10 mM BSA, 0.01% Tween).

Protein purification

To purify BacA-His₆, *E. coli* Rosetta(DE3)pLysS (Invitrogen) was transformed with pEC86 and grown in LB medium at 37 °C to an OD₆₀₀ of 0.8. Isopropyl-β-D-1-thiogalactopyranoside (IPTG) was added to a final concentration of 0.5 mM and the incubation was continued for another 3 h. The cells were harvested by centrifugation for 15 min at 7,500 ×g and 4 °C and washed with buffer B2 (50 mM NaH₂PO₄, 300 mM NaCl, 10 mM imidazole, adjusted to pH 8.0 with NaOH). Subsequently, they were resuspended in buffer B2 containing 100 μg/mL phenylmethylsulfonyl fluoride, 10 μg/mL DNase I and 1 mM β-mercaptoethanol and lysed by three passages through a French press at 16,000 psi. After the removal of cell debris by centrifugation at 30,000 ×g for 30 min, the supernatant was applied onto a 5 mL HisTrap HP column (GE Healthcare) equilibrated with buffer B3 (50 mM NaH₂PO₄, 300 mM NaCl, 20 mM imidazole, adjusted to pH 8.0 with NaOH). The column was washed with 10 column volumes (CV) of the same buffer, and protein was eluted at a flow rate of 1 mL/min with a linear imidazole gradient obtained by mixing buffers B3 and B4 (50 mM NaH₂PO₄, 300 mM NaCl, 250 mM imidazole, adjusted to pH 8.0 with NaOH). Fractions containing high concentrations of BacA-His₆ were pooled and dialyzed against 3 L of buffer B5 (20 mM Tris/HCl pH 8.0, 10

mM NaCl, 1 mM β -mercaptoethanol) at 4 °C. After the removal of precipitates by centrifugation at 30,000 $\times$ g for 30 min, the protein solution was aliquotted and snap-frozen in liquid nitrogen. Aliquots were stored at -80 °C until further use.

To purify LmdC, the protein was first produced as a His₆-SUMO-LmdC fusion (Marblestone et al., 2006). To this end, *E. coli* Rosetta(DE3)pLysS cells carrying plasmid pLY015 were grown at 37 °C in 3 L of LB medium supplemented with ampicillin and chloramphenicol. At an OD₆₀₀ of 0.6, the culture was chilled to 18°C, and protein synthesis was induced by the addition of 1 mM IPTG prior to overnight incubation at 18 °C. The cultures were harvested by centrifugation at 10,000 $\times$ g for 20 min at 4 °C and washed with lysis buffer (50 mM Tris/HCl pH 8.0, 300 mM NaCl). Cells were resuspended in lysis buffer supplemented with 5 mM imidazole, 10 mg/mL DNase I and 100 mg/mL PMSF. After three passages through a French press (16,000 psi), the cell lysate was clarified by centrifugation (30,000 $\times$ g, 30 min, 4 °C). Protein was then purified using zinc-affinity chromatography using a 1 mL Zn-NTA column (Cube Biotech) equilibrated with lysis buffer containing 5 mM imidazole. Protein was eluted with a linear gradient of 5 to 250 mM imidazole in lysis buffer at a flow rate of 1 mL/min. Fractions containing high concentrations of His₆-SUMO-LmdC were pooled and dialyzed against 3 L of low-salt lysis buffer (20 mM Tris/HCl pH 7.6, 50 mM NaCl, 10% (v/v) glycerol). After the addition of Ulp1 protease (Marblestone et al., 2006) and dithiothreitol (1 mM), the protein was incubated for 4 h at 4 °C to cleave off the His₆-SUMO tag. The solution was centrifuged for 30 min at 38,000 $\times$ g and 4 °C to remove precipitates and then subjected to ion exchange chromatography using a 1 mL HiTrap Q HP column (Cytiva, USA) equilibrated with low-salt lysis buffer. His₆-SUMO passed the column in the flow-through, and LmdC was eluted with a linear gradient of 150-1000 mM NaCl in low-salt buffer. Fractions containing LmdC were concentrated in an Amicon Ultra-4 10K spin concentrator (MWCO 10,000; Merck, Germany). After the removal of precipitates by centrifugation at 30,000 $\times$ g for 30 min, LmdC was further purified by size exclusion chromatography (SEC) on a HighLoad 16/60 Superdex 200 pg column (GE Healthcare, USA) equilibrated with SEC buffer (20 mM Tris/HCl pH 7.4, 150 mM NaCl). Fractions containing pure protein were pooled and concentrated. After the removal of precipitates by centrifugation at 30,000 $\times$ g, the protein solution was snap-frozen in liquid N₂ and stored at -80 °C until further use.

Bioinformatic analysis

Protein similarity searches were performed with BLAST (Altschul et al., 1990), using the BLAST server of the National Institutes of Health (<https://blast.ncbi.nlm.nih.gov/Blast.cgi>). Transmembrane helices were predicted with DeepTMHMM (Hallgren et al., 2022), coiled-coil regions with PCOILS (Lupas, 1996). The positions of conserved functional domains were determined using the PFAM server (Mistry et al., 2021). AlphaFold2 (Jumper et al., 2021) and AlphaFold-Multimer (Evans et al., 2022), as implemented in the AlphaFold.ipynb notebook on Google Colab, were used to predict the tertiary or quaternary structure of proteins, respectively. SuperPlots (Lord et al., 2020) were used to visualize cell length distributions and to evaluate the statistical significance of differences between multiple distributions, employing the PlotsOfData web app (Postma and Goedhart, 2019).

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Data availability

All relevant data generated in this study are included in the manuscript and the supplemental information.

Author contributions

M.O.V., E.C. and M.T. conceived the study. M.O.V., E.C. and S.P. constructed plasmids and strains and performed the growth analyses and subcellular localization studies. M.O.V. purified proteins and conducted the biochemical studies. P.S. analyzed the co-conservation of BacA and LmdC homologs. J.B. conducted the peptidoglycan analysis. R.H.-T. performed the single-particle tracking analysis. M.O.V., E.C., S.P. R.H.-T., P.S., J.B., W.V. and M.T. analyzed the data. W.V., P.L.G. and M.T. supervised the study. W.V., P.L.G. and M.T. secured funding. M.O.V. and M.T. wrote the paper, with input from all other authors.

Competing interests

The authors declare no competing interests.

Figure legends

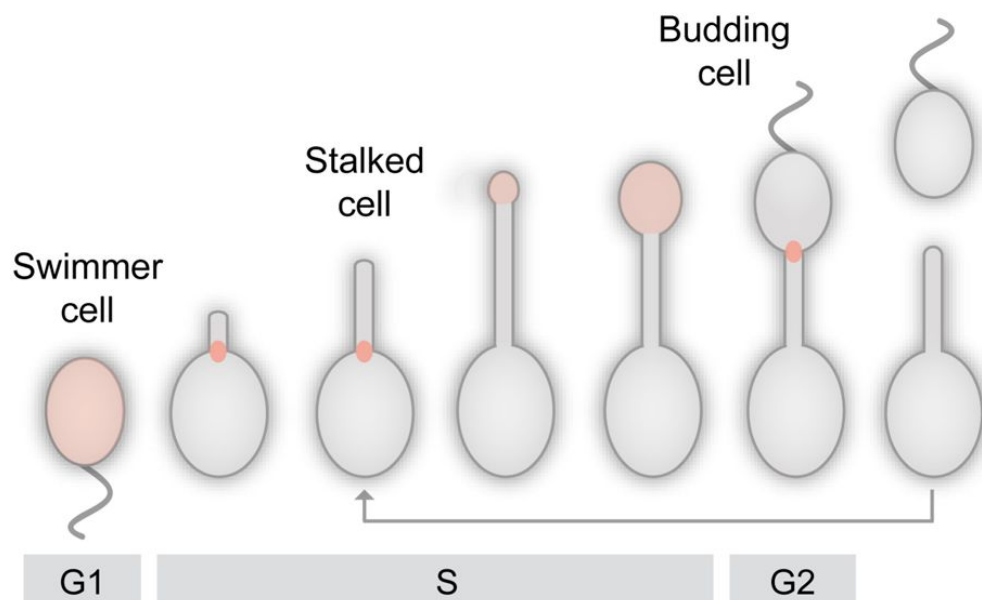


Figure 1—figure supplement 1.

The dimorphic lifecycle of *H. neptunium*.

A motile, flagellated swimmer sheds its flagellum and forms a stalk at the opposite cell pole. At a defined point in the cell cycle, the terminal segment of the stalk dilates and develops into a new swimmer cell. After cell division, the newborn swimmer cell first needs to differentiate into a stalked cell to initiate a new

round of cell division, whereas the stalked mother cell immediately enters the next replication cycle. The predominant growth zones (Cserti et al., 2017) are indicated in red.

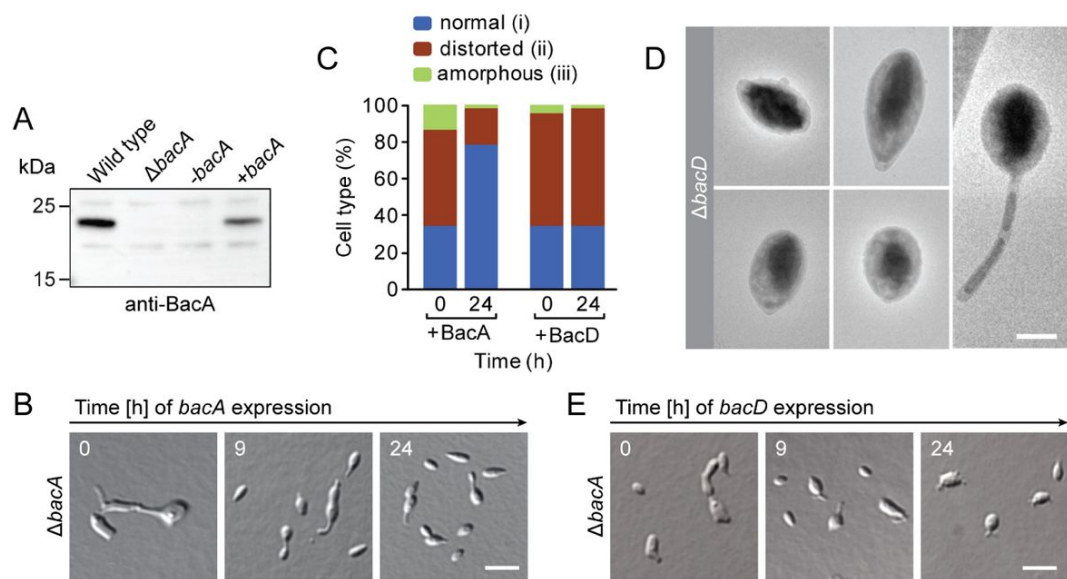


Figure 1—figure supplement 2.

Phenotypic analysis of *H. neptunium* bactofilin mutants.

(A) Immunoblot analysis of strains LE670 (wild type), EC28 ($\Delta bacA$) and EC41 ($\Delta bacA$ $P_{Cu}::P_{Cu}-bacA$) grown in the absence ($-bacA$) and presence ($+bacA$) of 0.5 mM $CuSO_4$, performed with an anti-BacA antibody. **(B)** Rescue of the phenotype of a $\Delta bacA$ mutant by ectopic expression of *bacA*. Cells of strain EC41 ($\Delta bacA$ $P_{Cu}::P_{Cu}-bacA$) were grown in copper-free medium, induced by the addition of copper, and imaged after the indicated time incubation times. Bar: 3 μm . **(C)** Quantification of the proportion of phenotypically abnormal stalked and budding cells in the cultures of strains EC41 ($\Delta bacA$ $P_{Cu}::P_{Cu}-bacA$) and EC43 ($\Delta bacA$ $P_{Zn}::P_{Zn}-bacD$) analyzed in panels B and E before ($t=0$ h) and 24 h after induction ($n = 100$ cells per time point). **(D)** Transmission electron micrographs of $\Delta bacD$ cells at the swimmer and stalk cell stage. Bar: 1 μm . **(E)** DIC images of a $\Delta bacA$ mutant overproducing BacD from a zinc-inducible promoter. Cells of strain EC43 ($\Delta bacA$ $P_{Zn}::P_{Zn}-bacD$) were grown in inducer-free medium, induced by the addition of 0.5 mM $ZnSO_4$, and imaged after the indicated incubation times. Bar: 3 μm .

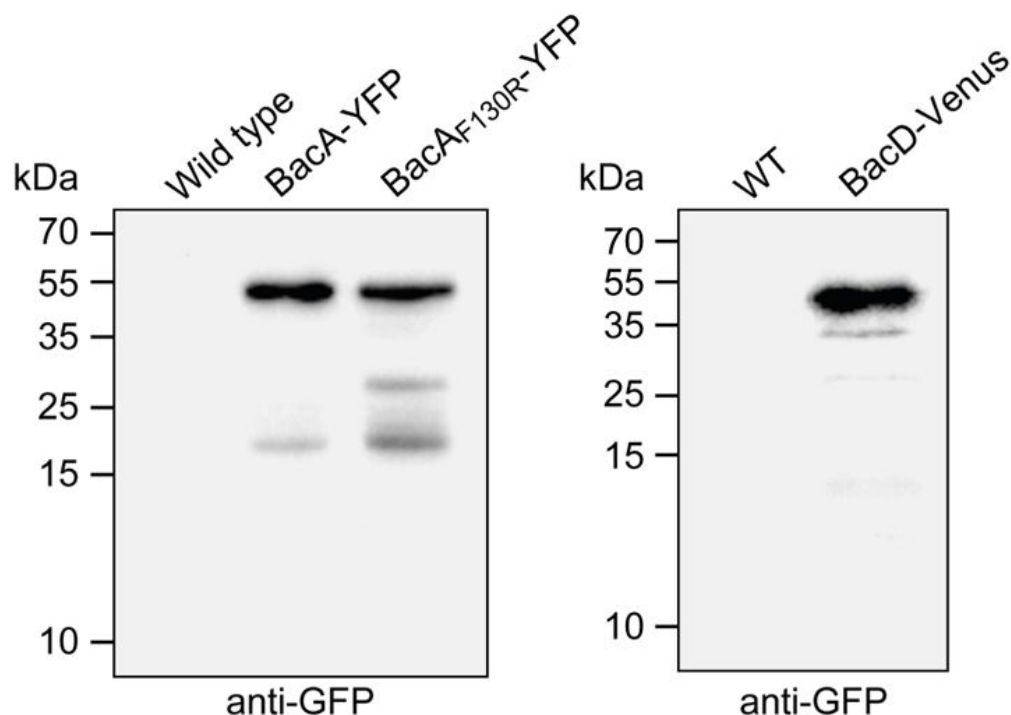


Figure 4—figure supplement 1.

Stability of the fluorescent protein fusions used in this study.

Cells of strains LE670 (wild type), EC61 (*bacA::bacA-eyfp*), MO78 (*bacA::bacAF130R-eyfp*) and EC67 (*bacD::bacD-venus*) were subjected to immunoblot analysis using anti-GFP antibodies.

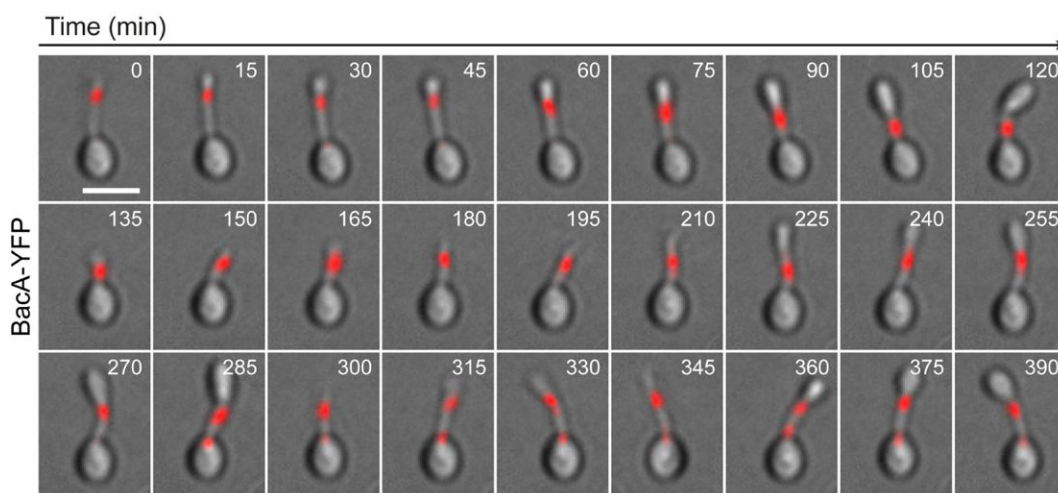


Figure 4—figure supplement 2.

Localization dynamics of BacA-YFP.

Cells of strain EC61 (*bacA::bacA-eyfp*) were flushed into a microfluidic flow cell and imaged at 15-min intervals over multiple cell cycles. Shown are overlays of DIC and fluorescence images. Bar: 2 μm.

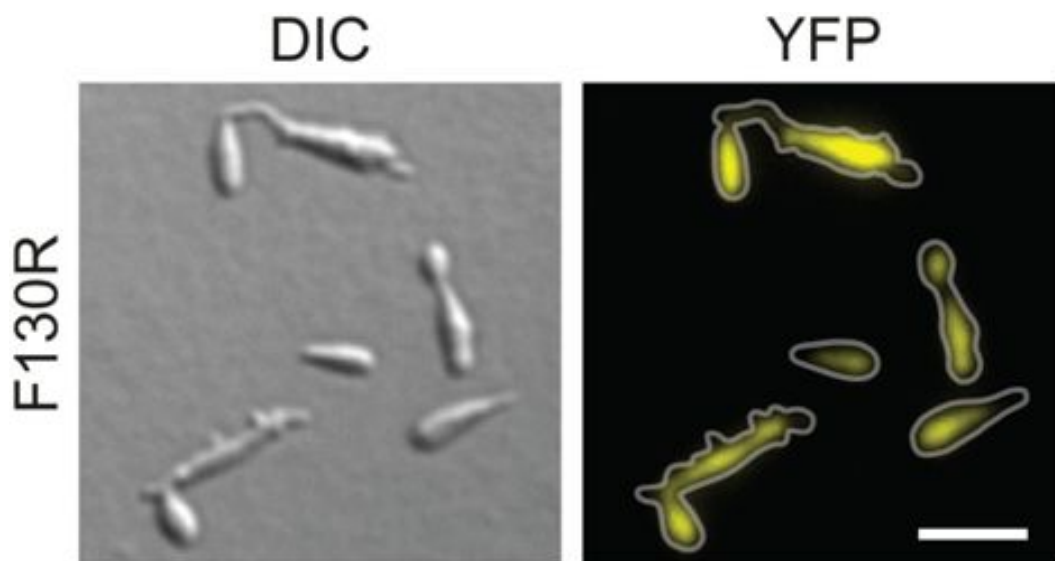
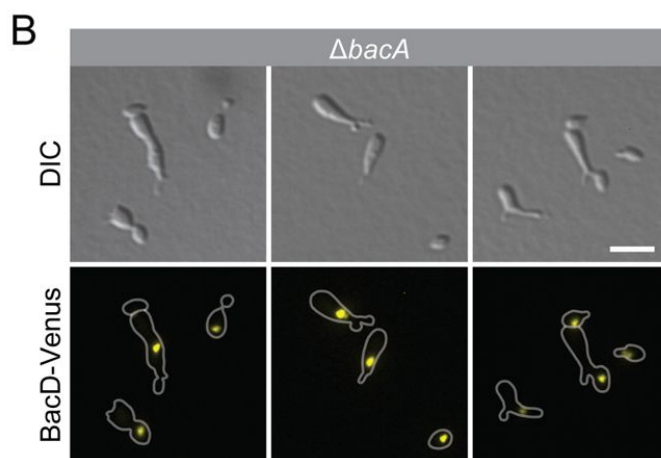
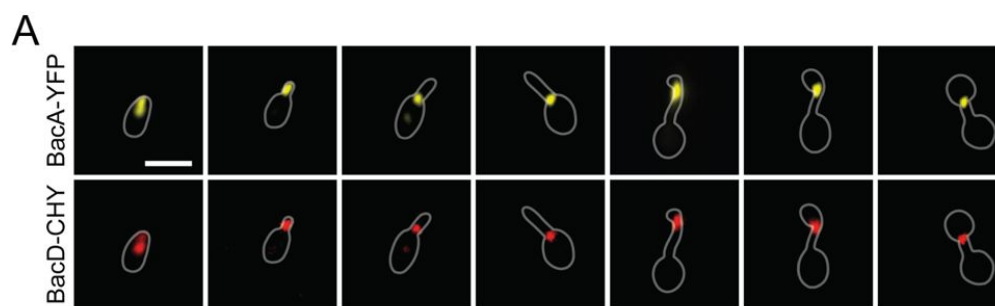


Figure 4—figure supplement 3.

Dependence of BacD localization on BacA.

(A) Co-localization of BacA and BacD in cells of strain EC68 (*bacA::bacA-eyfp bacD::bacD-mCherry*). Shown are the individual fluorescence images used to generate the overlays in Figure 4D. (B) Random localization of BacD complexes in the absence of BacA. Cells of strain EC60 ($\Delta bacA$ $P_{Zn}::P_{Zn}-bacD-venus$) were grown in inducer-free medium and induced for 4 h with 0.5 mM $ZnSO_4$ prior to imaging. Bar: 3 μm .

Figure 4—figure supplement 4.

Localization and functionality of BacA_{F130R}-YFP.

Cells of strain MO78 (*bacA::bacA_{F130R}-eyfp*), producing the polymerization-deficient BacA_{F130R}-YFP variant instead of the native protein, were grown to exponential phase and imaged. Bar: 3 μm .

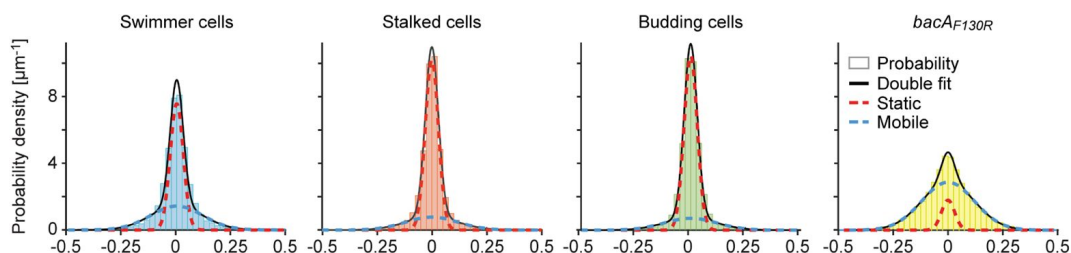


Figure 5—figure supplement 1.

Single-particle tracking analysis of BacA-YFP mobility.

Shown is a Gaussian mixture model (GMM) analysis of the mobility of the BacA-YFP variants analyzed in Figure 3B. The distributions of the frame-to-frame displacements in both x and y direction from single-particle tracking experiments were fitted to a two-component Gaussian function (sum in black), assuming a fast-diffusing mobile (dotted blue lines) and a slow-diffusing (dotted red lines) population.

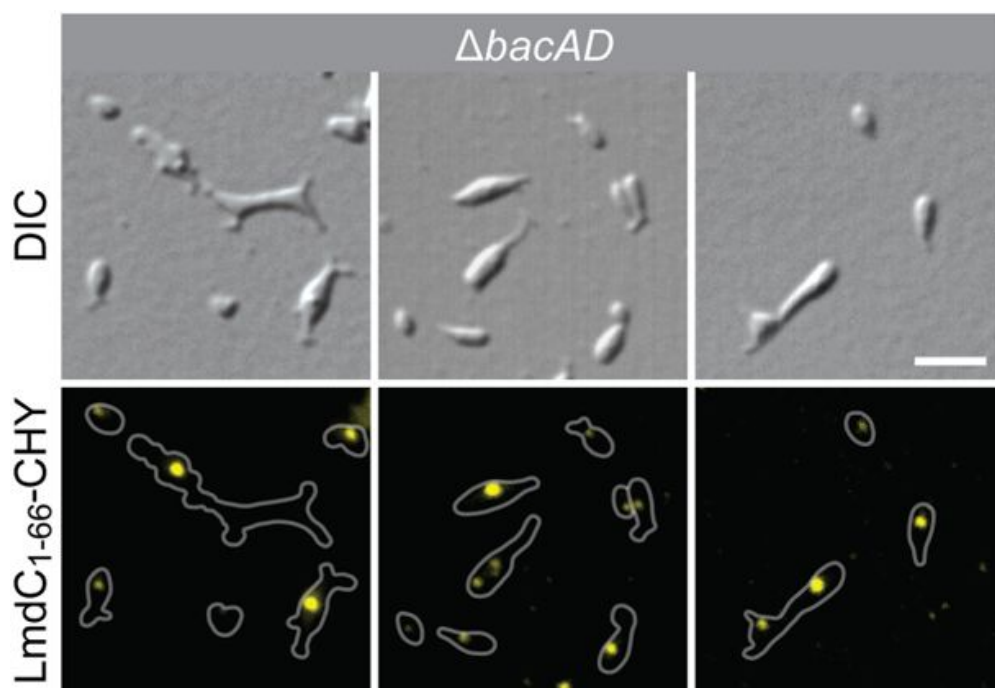


Figure 8—figure supplement 1.

Random localization of LmdC₁₋₆₆-mCherry in the $\Delta bacAD$ background.

Cells of strain MO80 ($\Delta bacAD$ $P_{Cu}::P_{Cu}$ -LmdC₁₋₆₆-mCherry) were induced for 2 h with 0.5 mM CuSO₄ prior to imaging. mCherry fluorescence is shown in yellow. Bar: 3 μ m.

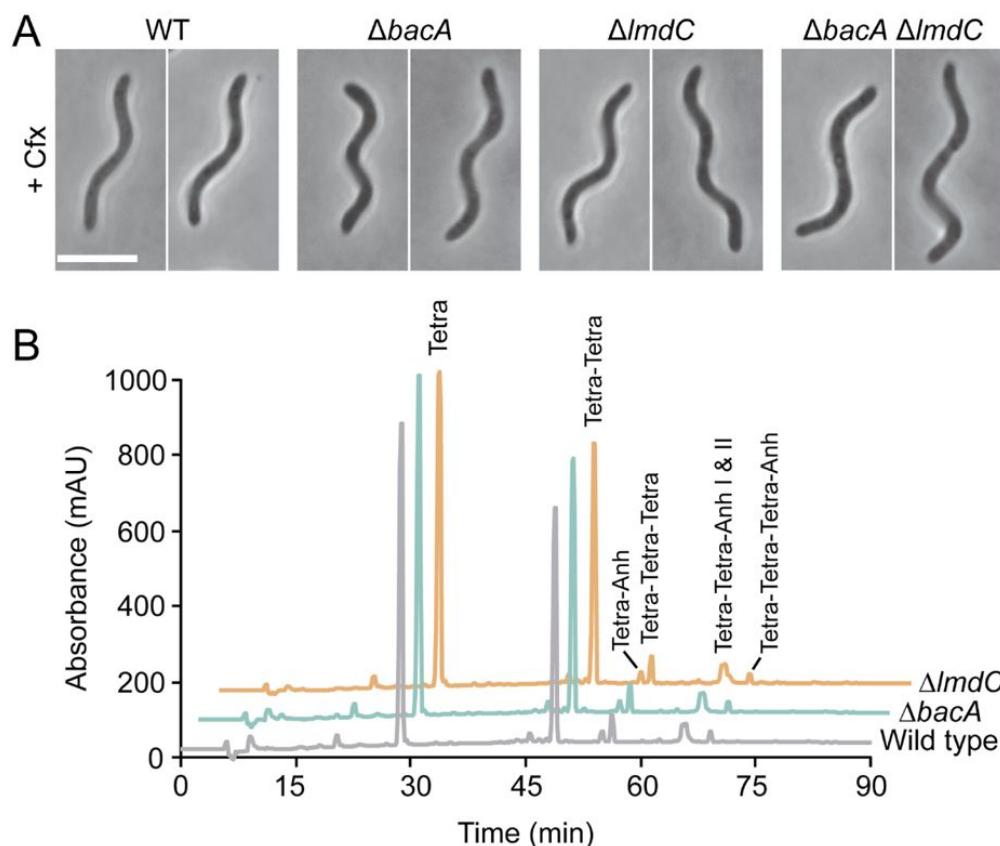


Figure 9—figure supplement 1.

Analysis of BacA and LmdC from *R. rubrum*.

(A) Clearer visualization of the abnormal curvature of BacA- and LmdC-deficient *R. rubrum* cells by inhibition of cell division. Cells of strains SP68 ($\Delta lmdC$), SP70 ($\Delta bacA$) and SP116 ($\Delta lmdC \Delta bacA$) were treated with 5 μ g/ml cefalexin (Cfx) for 6 h. Bar: 5 μ m. (B) Muropeptide profiles of different *R. rubrum* strains. Cell walls of strains S1 (wild type), SP70 ($\Delta bacA_{RS}$) and SP68 ($\Delta lmdC$) were digested with cellosyl to release muropeptides, which were reduced and separated by HPLC chromatography. The identities of major

muropeptide species, assigned based on the previously reported retention times (Glauner, 1988), are given above the corresponding peaks. Tetra stands for N-acetylglucosamine-N-acetylmuramitol tetrapeptide. "Anh" indicates muropeptides containing 1,6-anhydromuramic acid.

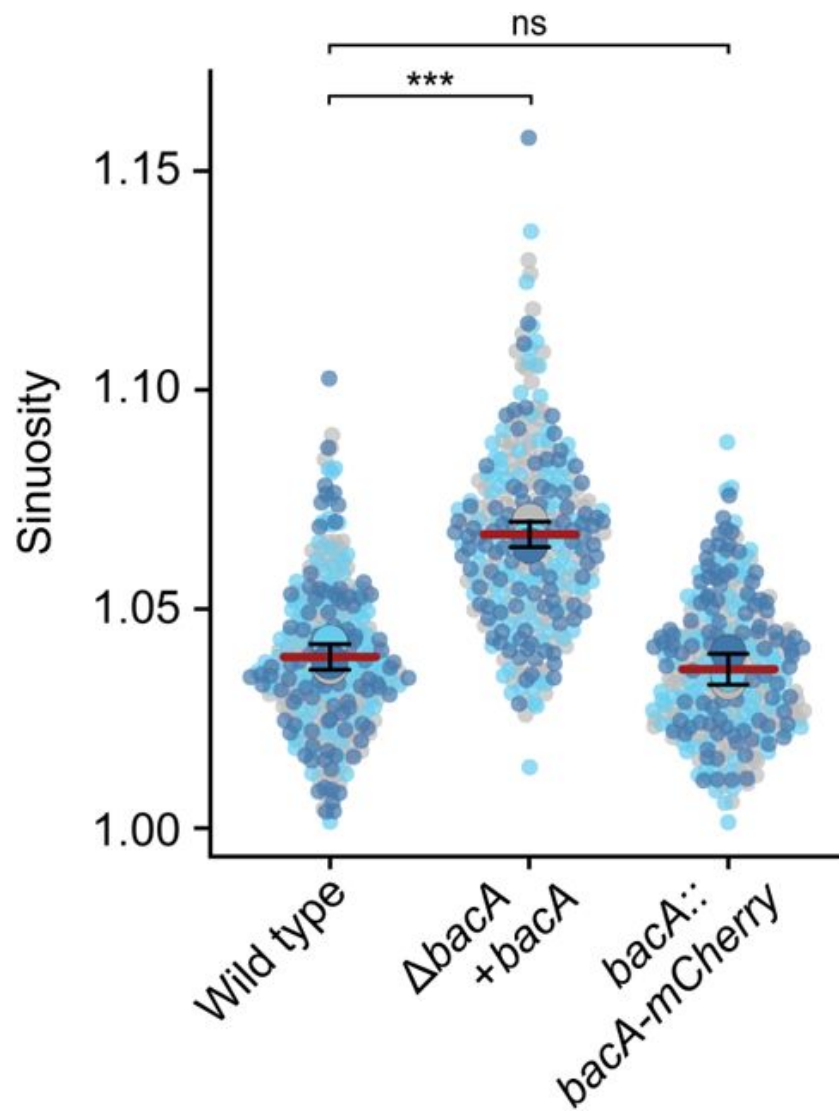


Figure 9—figure supplement 2.

Cell lengths of *R. rubrum* strains.

Superplots showing the distribution of cell sinuosities in populations of strains S1 (wild type), SP105 ($\Delta bacA_{RS}$ P_{bacA} - $bacA_{RS}$) and SP109 ($bacA_{RS}::bacA_{RS}$ -*mCherry*). Small dots represent the data points, large dots represent the median values of the three independent experiments shown in the graphs (dark blue, light blue, grey). The mean of the three median values is indicated by a horizontal line. Whiskers represent the standard deviation. *** $p < 0.005$; ns, not significant; t-test).

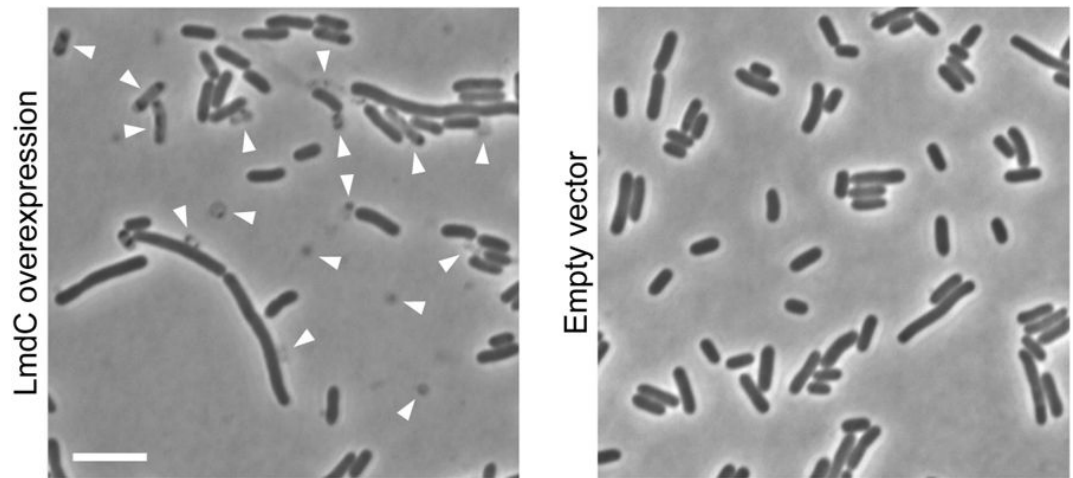


Figure 9—figure supplement 3.

Lysis of *E. coli* upon heterologous expression of full-length *R. rubrum lmdC*.

Shown are cells of *E. coli* Rosetta(DE3)pLysS harboring pSP120 (P_{T7} -*lmdC*_{RS}) or the corresponding empty vector imaged 4h after the induction of gene expression with 0.5 mM IPTG. Arrowheads indicate ghost cells, cell debris, lysing cells or strongly elongated cells. Scale bar: 5 μ m.

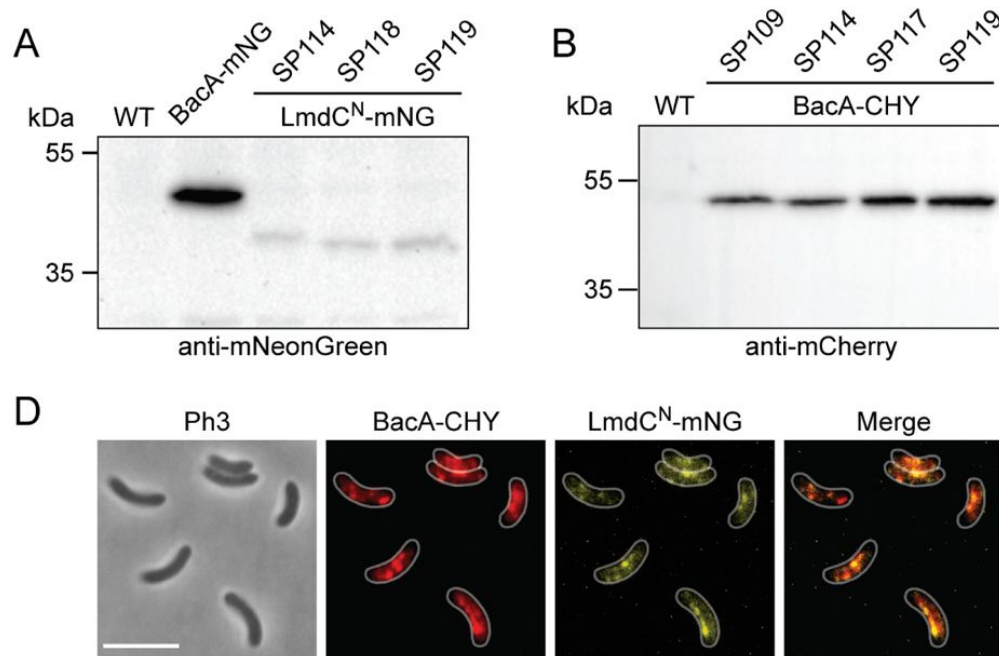


Figure 9—figure supplement 4.

Analysis of *R. rubrum* strains producing fluorescent protein fusions.

(A) Levels of the LmdC^N_{RS}-mNeonGreen fusion in different *R. rubrum* strains. Strains SP98 (Δ lmdC_{RS} bacA_{RS}::bacA_{RS}-mNeongreen), SP114 (bacA_{RS}::bacA_{RS}-mCherry P_{lmdC}-lmdC^N_{RS}-mNeongreen), SP118 (Δ bacA Δ lmdC P_{lmdC}-lmdC₁₋₈₀-mNeongreen) and SP119 (bacA::bacA-mCherry Δ lmdC

P_{lmdC}-lmdC₁₋₈₀-mNeongreen) were subjected to immunoblot analysis with an anti-mNeonGreen antibody. Predicted molecular weights: BacA_{RS}-mNeonGreen (44.8 kDa), LmdC^N_{RS}-mNeonGreen fusion (35.9 kDa) **(B)** Levels of the BacA_{RS}-mCherry fusion in different *R. rubrum* strains. Strains SP109 (bacA_{RS}::bacA_{RS}-mCherry), SP114 (bacA_{RS}::bacA_{RS}-mCherry P_{lmdC}-lmdC^N_{RS}-mNeongreen), SP117 (bacA::bacA-mCherry Δ lmdC) and SP119 (bacA::bacA-mCherry Δ lmdC P_{lmdC}-lmdC^N_{RS}-mNeongreen) were subjected to immunoblot analysis with an anti-mCherry antibody. The predicted molecular weight of BacA_{RS}-mCherry is 44.9 kD. **(C)** Co-localization of BacA_{RS}-mCherry and LmdC_{RS}-mNeongreen in *R. rubrum* cells carrying the native lmdC_{RS} gene. The Pearson's Correlation Coefficient (PCC) of the two fluorescence signals in a random subpopulation of cells (n=177) is 0.54. Scale bar: 5 μ m.

Video legends

Video 1. Unregulated growth of a bactofilin-deficient *H. neptunium* mutant. Cells of strain EC33 (Δ badAD) were grown in a microfluidic flow cell and imaged by DIC microscopy at 15 min intervals. Bar: 2 μ m.

Video 2. Normal growth of the *H. neptunium* wild-type strain. Cells of strain LE670 (wild type) were grown in a microfluidic flow cell and imaged by DIC microscopy at 5 min intervals. Bar: 2 μ m.

Videos 3. Localization dynamics of BacA-YFP (example 1). Cells of strain EC61 (bacA::bacA-eyfp) were grown in a microfluidic flow cell and imaged at 15 min intervals. Shown are overlays of DIC and fluorescence images. YFP fluorescence is shown in red for better visibility. Bar: 2 μ m.

Videos 4. Localization dynamics of BacA-YFP (example 2). Images were taken as described for Video 3. Bar: 2 μ m.

Videos 5. Localization dynamics of BacA-YFP (example 3). Images were taken as described for Video 3. Bar: 2 μ m.

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Author information

1. Manuel Osorio-Valeriano

Department of Biology, University of Marburg, Marburg, Germany, Max Planck Institute for Terrestrial Microbiology, Marburg, Germany

2. Emöke Cserti

Department of Biology, University of Marburg, Marburg, Germany

3. Sebastian Pöhl

Department of Biology, University of Marburg, Marburg, Germany

4. Rogelio Hernández-Tamayo

Department of Chemistry, University of Marburg, Marburg, Germany, Center for Synthetic Microbiology (SYNMIKRO), Marburg, Germany

5. Jacob Biboy

Centre for Bacterial Cell Biology, Institute for Cell and Molecular Biosciences, Newcastle University, Newcastle upon Tyne, United Kingdom
ORCID iD: [0000-0002-1286-6851](https://orcid.org/0000-0002-1286-6851)

6. Patrick Sobetzko

Center for Synthetic Microbiology (SYNMIKRO), Marburg, Germany

7. Waldemar Vollmer

Centre for Bacterial Cell Biology, Institute for Cell and Molecular Biosciences, Newcastle University, Newcastle upon Tyne, United Kingdom
ORCID iD: [0000-0003-0408-8567](https://orcid.org/0000-0003-0408-8567)

8. Peter L. Graumann

Department of Chemistry, University of Marburg, Marburg, Germany, Center for Synthetic Microbiology (SYNMIKRO), Marburg, Germany

9. Martin Thanbichler

Department of Biology, University of Marburg, Marburg, Germany, Max Planck Institute for Terrestrial Microbiology, Marburg, Germany, Center for Synthetic Microbiology (SYNMIKRO), Marburg, Germany

For correspondence:

thanbichler@uni-marburg.de
ORCID iD: [0000-0002-1303-1442](https://orcid.org/0000-0002-1303-1442)

Editors

Reviewing Editor

Sonja Albers

University of Freiburg, Germany

Senior Editor

Dominique Soldati-Favre

University of Geneva, Switzerland

Reviewer #1 (Public Review):

In their study, Osorio-Valeriano and colleagues seek to understand how bacterial-specific polymerizing proteins called bactofilins contribute to morphogenesis. They do this primarily in the stalked budding bacterium *Hyphomonas neptunium*, with supporting work in a spiral-shaped bacterium, *Rhodospirillum rubrum*. Overall the study incorporates bacterial genetics and physiology, imaging, and biochemistry to explore the function of bactofilins and cell wall hydrolases that are frequently encoded together within an operon. They demonstrate an important, but not essential, function for BacA in morphogenesis of *H. neptunium*. Using biochemistry and imaging, they show that BacA can polymerize and that its localization in cells is dynamic and cell-cycle regulated. The authors then focus on LmdC, which encodes a putative M23 endopeptidase upstream of *bacA* in *H. neptunium*, and find that is essential for viability. The purified LmdC C-terminal domain could cleave *E. coli* peptidoglycan in vitro suggesting that it is a DD-endopeptidase. LmdC interacts directly with BacA in vitro and co-localizes with BacA in cells. To expand their observations, the authors then explore a related endopeptidase/bactofilin pair in *R. rubrum*; those observations support a function for LmdC and BacA in *R. rubrum* morphogenesis as well.

An overall strength of this study is the breadth and completeness of approaches used to assess bactofilin and endopeptidase function in cells and in vitro. The authors establish a clear function for BacA in morphogenesis in two bacterial systems, and demonstrate a physical relationship between BacA and the cell wall hydrolase LmdC that may be broadly conserved. The eventual model the authors favor for BacA regulation of morphogenesis in *H. neptunium* is that it serves as a diffusion barrier and limits movement of morphogenetic machinery like the elongasome into the elongating stalk and/or bud. However, there is no data presented here to address that model and the role of LmdC in *H. neptunium* morphogenesis remains unclear.

The data presented illuminate aspects of bacterial morphogenesis and the physical and functional relationship between polymerizing proteins and cell wall enzymes in bacteria, a recurring theme in bacterial cell biology with a variety of underlying mechanisms. Bactofilins in particular are relatively recently discovered and any new insights into their functions and mechanisms of action are valuable. The findings presented here are likely to interest those studying bacterial morphogenesis, peptidoglycan, and cytoskeletal function.

Reviewer #2 (Public Review):

This is an excellent study. It starts with the identification of two bactofilins in *H. neptunium*, a demonstration of their important role for the determination of cell shape and discovery of an associated endopeptidase to provide a convincing model for how these two classes of

proteins interact to control cell shape. This model is backed up by a quantitative characterisation of their properties using high-resolution imaging and image analysis methods.

Overall, all evidence is very convincing and I do not have many recommendations on how to improve the manuscript.

In my opinion, there are only two issues that I have with the paper:

1. The single particle dynamics of BacA is presented as analysed and I would like to give some suggestions how to maybe extract even more information from the already acquired data:

1.1. Presentation: Figure 5A is only showing projections of single particle time-lapse movies. To convince the reader that it was indeed possible to detect single molecules it would be helpful if the authors present individual snapshots and intensity traces. In case of single molecules these will show step wise bleaching.

1.2. Analysis: Figure 5B and Supplement Figure 1 are showing the single particle tracking results, revealing that there are two populations of BacA-YFP in the cell. However, this data does not show if individual BacA particles transition between these two populations or not. A more detailed analysis of the existing data, where one can try to identify confinement events in single particle trajectories could be very revealing and help to understand the behaviour of BacA in more detail.

2. The title of Fig. 3 says that BacA and BacD copolymerise, however, the data presented to confirm this conclusion is actually rather weak. First, the AlphaFold prediction does not show the co-polymer, and second, the in vitro polymerisation experiments were only done with BacA in the absence of BacD. Accordingly, the only evidence that supports this is their colocalization in fluorescence microscopy. I suggest either weakening the statement or changing the title adds more evidence.

Finally, did the authors think about biochemical experiments to study the interaction between the cytoplasmic part of LmdC and the bactofilins? These could further support their model.